

# TL06xx Low-Power JFET-Input Operational Amplifiers

## 1 Features

- Very low power consumption
- Typical supply current: 200  $\mu$ A (per amplifier)
- Wide common-mode and differential voltage ranges
- Low input bias and offset currents
- Common-mode input voltage range includes  $V_{CC+}$
- Output short-circuit protection
- High input impedance: JFET-input stage
- Internal frequency compensation
- Latch-up-free operation
- High slew rate: 3.5 V/ $\mu$ s typical
- On products compliant to MIL-PRF-38535, all parameters are tested unless otherwise noted. On all other products, production processing does not necessarily include testing of all parameters.



## 2 Applications

- [Tablets](#)
- [White goods](#)
- [Personal electronics](#)
- [Computers](#)

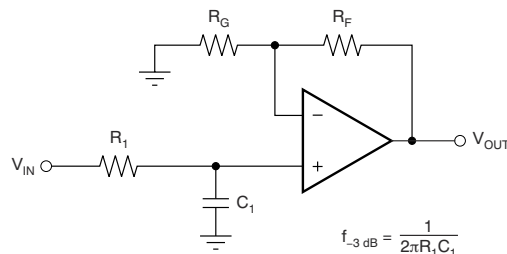
## 3 Description

The TL06x (TL061, TL062, and TL064) family of industry-standard operational amplifiers (op amps) mirror the TL07x and TL08x family of op amps with lower power consumption. These devices provide outstanding value for cost-sensitive applications, featuring high input impedance, wide bandwidth, high slew rate, and low input offset and input bias currents. High ESD (1.5 kV, HBM), integrated EMI and RF filters, and wide temperature operation enable the TL06x devices to be used in rugged and environmentally-demanding applications.

### Device Information

| PART NUMBER | CHANNEL COUNT | PACKAGE <sup>(1)</sup> | PACKAGE SIZE <sup>(2)</sup> |
|-------------|---------------|------------------------|-----------------------------|
| TL061x      | Single        | D (SOIC, 8)            | 4.90 mm × 6.00 mm           |
|             |               | P (PDIP, 8)            | 9.59 mm × 7.94 mm           |
|             |               | PS (SO, 8)             | 6.20 mm × 7.80 mm           |
| TL062x      | Dual          | D (SOIC, 8)            | 4.90 mm × 6.00 mm           |
|             |               | P (PDIP, 8)            | 9.59 mm × 7.94 mm           |
|             |               | PS (SO, 8)             | 6.20 mm × 7.80 mm           |
|             |               | JG (CDIP, 8)           | 9.58 mm × 7.62 mm           |
|             |               | PW (TSSOP, 8)          | 3.00 mm × 6.40 mm           |
|             |               | FK (LCCC, 20)          | 8.89 mm × 8.80 mm           |
| TL064x      | Quad          | D (SOIC, 14)           | 8.65 mm × 6.00 mm           |
|             |               | J (CDIP, 14)           | 19.4 mm × 7.90 mm           |
|             |               | N (PDIP, 14)           | 19.31 mm × 7.94 mm          |
|             |               | NS (SO, 14)            | 10.20 mm × 7.80 mm          |
|             |               | PW (TSSOP, 14)         | 5.00 mm × 6.40 mm           |
|             |               | W (CFP, 14)            | 21.78 mm × 9.21 mm          |
|             |               | FK (LCCC, 20)          | 8.89 mm × 8.80 mm           |

- (1) For all available packages, see the orderable addendum at the end of the data sheet.
- (2) The package size (length × width) is a nominal value and includes pins, where applicable.



$$\frac{V_{OUT}}{V_{IN}} = \left(1 + \frac{R_F}{R_G}\right) \left(\frac{1}{1 + sR_1 C_1}\right)$$

**Single-Pole, Low-Pass Filter**



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## 4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

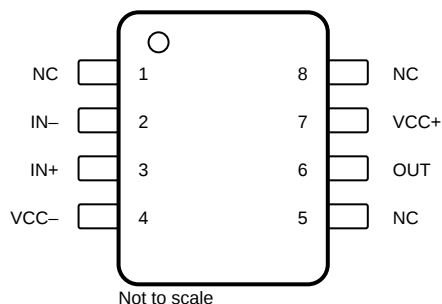
| Changes from Revision M (June 2023) to Revision N (August 2023)                                              | Page |
|--------------------------------------------------------------------------------------------------------------|------|
| • Added typical specification for Unity-Gain Bandwidth in <i>Electrical Characteristics for TL06xM</i> ..... | 10   |
| • Changed Equivalent Input Noise Voltage vs Frequency curve in <i>Typical Characteristics</i> section.....   | 11   |

| Changes from Revision L (May 2015) to Revision M (June 2023)                                                                             | Page |
|------------------------------------------------------------------------------------------------------------------------------------------|------|
| • Updated the numbering format for tables, figures, and cross-references throughout the document.....                                    | 1    |
| • Updated <i>Device Information</i> with package size and channel count, and reordered packages based on<br>channel count.....           | 1    |
| • Updated TL061 pinout diagram in <i>Pin Configuration and Functions</i> .....                                                           | 4    |
| • Changed Charged Device Model (CDM) ESD from 2 kV to 1.5 kV in <i>ESD Ratings</i> .....                                                 | 6    |
| • Added table note for input bias current and input offset current on <i>Electrical Characteristics for TL06xC and<br/>TL06xxC</i> ..... | 8    |
| • Added table note for input bias current and input offset current on <i>Electrical Characteristics for TL06xxC and<br/>TL06xI</i> ..... | 9    |
| • Changed name of <i>Electrical Characteristics for TL06xM and TL064M</i> to <i>Electrical Characteristics for TL06xM</i><br>.....       | 10   |
| • Added table note for input bias current and input offset current on <i>Electrical Characteristics for TL06xM</i> .....                 | 10   |
| • Changed typical input voltage noise density at 1 kHz from 42 nV/√Hz to 30 nV/√Hz .....                                                 | 10   |
| • Updated description in <i>Overview</i> .....                                                                                           | 15   |
| • Updated image in <i>Functional Block Diagram</i> .....                                                                                 | 15   |

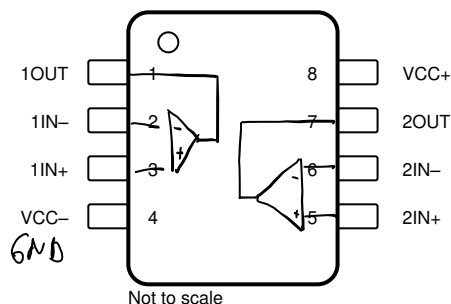
| Changes from Revision K (January 2014) to Revision L (May 2015)                                                                                                                                                                                                                                                                                                                                                | Page |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| • Added <i>Applications</i> .....                                                                                                                                                                                                                                                                                                                                                                              | 1    |
| • Added <i>Pin Configuration and Functions</i> section, <i>ESD Ratings</i> table, <i>Feature Description</i> section, <i>Device<br/>Functional Modes</i> , <i>Application and Implementation</i> section, <i>Power Supply Recommendations</i> section, <i>Layout</i><br>section, <i>Device and Documentation Support</i> section, and <i>Mechanical, Packaging, and Orderable Information</i><br>section ..... | 1    |

| <b>Changes from Revision J (September 2004) to Revision K (January 2014)</b>   | <b>Page</b>       |
|--------------------------------------------------------------------------------|-------------------|
| • Updated document to new TI data sheet format - no specification changes..... | <a href="#">1</a> |
| • Deleted <i>Ordering Information</i> table.....                               | <a href="#">1</a> |
| • Updated <i>Features</i> with Military Disclaimer.....                        | <a href="#">1</a> |

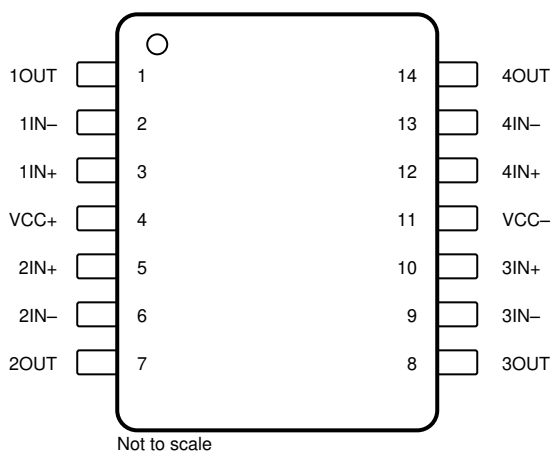
## 5 Pin Configuration and Functions



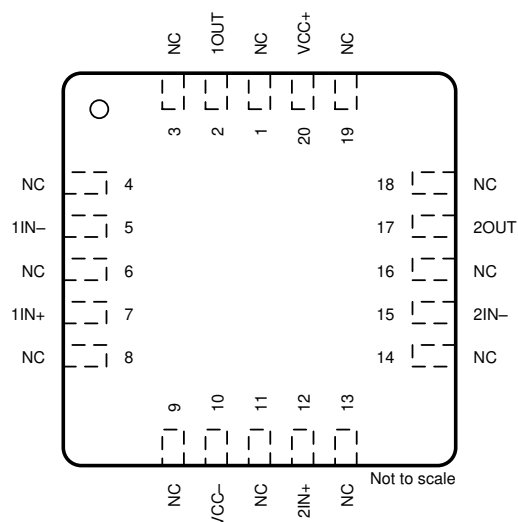
**Figure 5-1. TL061x D, P, and PS Package, 8-Pin SOIC, PDIP, and SO (Top View)**



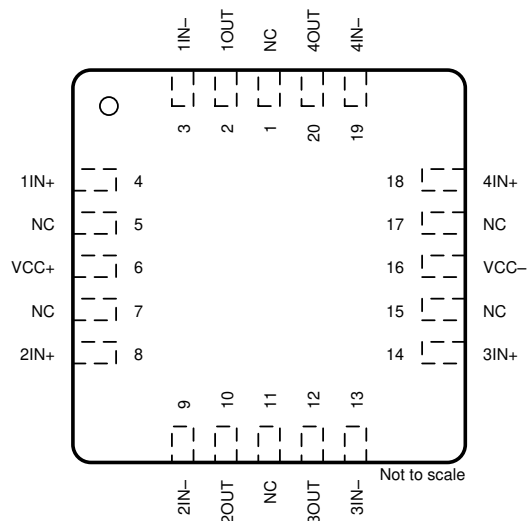
**Figure 5-2. TL062x D, JG, P, PS, and PW Package, 8-Pin SOIC, CDIP, PDIP, SO, and TSSOP (Top View)**



**Figure 5-3. TL064x D, J, N, NS, PW, and W Package, 14-Pin SOIC, CDIP, PDIP, SO, TSSOP and CFP (Top View)**



**Figure 5-4. TL062 FK Package, 20-Pin LCCC (Top View)**



**Figure 5-5. TL064 FK Package, 20-Pin LCCC (Top View)**

**Table 5-1. Pin Functions**

| PIN              |          |                  |    |                    |    | TYPE <sup>(1)</sup> | DESCRIPTION             |
|------------------|----------|------------------|----|--------------------|----|---------------------|-------------------------|
| NAME             | TL061    | TL062            |    | TL064              |    |                     |                         |
|                  | D, P, PS | D, JG, P, PS, PW | FK | D, J, N, NS, PW, W | FK |                     |                         |
| 1IN–             | —        | 2                | 5  | 2                  | 3  | I                   | Negative input          |
| 1IN+             | —        | 3                | 7  | 3                  | 4  | I                   | Positive input          |
| 1OUT             | —        | 1                | 2  | 1                  | 2  | O                   | Output                  |
| 2IN–             | —        | 6                | 15 | 6                  | 9  | I                   | Negative input          |
| 2IN+             | —        | 5                | 12 | 5                  | 8  | I                   | Positive input          |
| 2OUT             | —        | 7                | 17 | 7                  | 10 | O                   | Output                  |
| 3IN–             | —        | —                | —  | 9                  | 13 | I                   | Negative input          |
| 3IN+             | —        | —                | —  | 10                 | 14 | I                   | Positive input          |
| 3OUT             | —        | —                | —  | 8                  | 12 | O                   | Output                  |
| 4IN–             | —        | —                | —  | 13                 | 19 | I                   | Negative input          |
| 4IN+             | —        | —                | —  | 12                 | 18 | I                   | Positive input          |
| 4OUT             | —        | —                | —  | 14                 | 20 | O                   | Output                  |
| IN–              | 2        | —                | —  | —                  | —  | I                   | Negative input          |
| IN+              | 3        | —                | —  | —                  | —  | I                   | Positive input          |
| NC               | 8        | —                | 1  | —                  | 1  | —                   | Do not connect          |
|                  |          |                  | 3  |                    | 5  |                     |                         |
|                  |          |                  | 4  |                    |    |                     |                         |
|                  |          |                  | 6  |                    | 7  |                     |                         |
|                  |          |                  | 8  |                    |    |                     |                         |
|                  |          |                  | 9  |                    | 11 |                     |                         |
|                  |          |                  | 11 |                    |    |                     |                         |
|                  |          |                  | 13 |                    | 15 |                     |                         |
|                  |          |                  | 14 |                    |    |                     |                         |
|                  |          |                  | 16 |                    | 17 |                     |                         |
|                  |          |                  | 18 |                    |    |                     |                         |
|                  |          |                  | 19 |                    |    |                     |                         |
| OFFSET N1        | 1        | —                | —  | —                  | —  | —                   | Input offset adjustment |
| OFFSET N2        | 5        | —                | —  | —                  | —  | —                   | Input offset adjustment |
| OUT              | 6        | —                | —  | —                  | —  | O                   | Output                  |
| V <sub>CC–</sub> | 4        | 4                | 10 | 11                 | 16 | —                   | Power supply            |
| V <sub>CC+</sub> | 7        | 8                | 20 | 4                  | 6  | —                   | Power supply            |

(1) I = input, O = output

## 6 Specifications

### 6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)<sup>(1)</sup>

|                  |                                                              | MIN                            | MAX       | UNIT |
|------------------|--------------------------------------------------------------|--------------------------------|-----------|------|
| V <sub>CC+</sub> | Supply voltage <sup>(2)</sup>                                |                                | 18        | V    |
| V <sub>CC-</sub> |                                                              |                                | -18       |      |
| V <sub>ID</sub>  | Differential input voltage <sup>(3)</sup>                    |                                | ±30       | V    |
| V <sub>I</sub>   | Input voltage <sup>(2) (4)</sup>                             |                                | ±15       | V    |
|                  | Duration of output short circuit <sup>(5)</sup>              |                                | Unlimited |      |
| T <sub>J</sub>   | Operating virtual junction temperature                       |                                | 150       | °C   |
|                  | Case temperature for 60 seconds                              | FK package                     | 260       | °C   |
|                  | Lead temperature 1.6 mm (1/16 inch) from case for 60 seconds | J, JG, U, or W package         | 300       | °C   |
|                  | Lead temperature 1.6 mm (1/16 inch) from case for 10 seconds | D, N, NS, P, PS, or PW package | 260       | °C   |
| T <sub>stg</sub> | Storage temperature                                          | -65                            | 150       | °C   |

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) All voltage values, except differential voltages, are with respect to the midpoint between V<sub>CC+</sub> and V<sub>CC-</sub>.
- (3) Differential voltages are at IN+, with respect to IN-.
- (4) The magnitude of the input voltage must never exceed the magnitude of the supply voltage or 15 V, whichever is less.
- (5) The output may be shorted to ground or to either supply. Temperature or supply voltages must be limited so that the dissipation rating is not exceeded.

### 6.2 ESD Ratings

|                    |                                                                                | VALUE | UNIT |
|--------------------|--------------------------------------------------------------------------------|-------|------|
| V <sub>(ESD)</sub> | Electrostatic discharge                                                        |       |      |
|                    | Human body model (HBM), per ANSI/ESDA/JEDEC JS-001 <sup>(1)</sup>              | 2000  | V    |
|                    | Charged-device model (CDM), per JEDEC specification JESD22-C101 <sup>(2)</sup> | 1500  |      |

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

### 6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

|                  |                     | MIN                  | MAX                  | UNIT |
|------------------|---------------------|----------------------|----------------------|------|
| V <sub>CC+</sub> | Supply voltage      | 5                    | 15                   | V    |
| V <sub>CC-</sub> | Supply voltage      | -5                   | -15                  | V    |
| V <sub>CM</sub>  | Common-mode voltage | V <sub>CC-</sub> + 4 | V <sub>CC+</sub> - 4 | V    |
| T <sub>A</sub>   | Ambient temperature |                      |                      | °C   |
|                  | TL06xM              | -55                  | 125                  |      |
|                  | TL06xQ              | -40                  | 125                  |      |
|                  | TL06xI              | -40                  | 85                   |      |
|                  | TL06xC              | 0                    | 70                   |      |

## 6.4 Thermal Information (TL061)

| THERMAL METRIC <sup>(1)</sup> |                                                              | TL061    |          | UNIT |
|-------------------------------|--------------------------------------------------------------|----------|----------|------|
|                               |                                                              | D (SOIC) | P (PDIP) |      |
|                               |                                                              | 8 PINS   | 8 PINS   |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance <sup>(2) (3)</sup>    | 97       | 85       | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance <sup>(4) (5)</sup> | —        | —        | °C/W |

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).
- (2) Maximum power dissipation is a function of T<sub>J(max)</sub>, R<sub>θJA</sub>, and T<sub>A</sub>. The maximum allowable power dissipation at any allowable ambient temperature is P<sub>D</sub> = (T<sub>J(max)</sub> – T<sub>A</sub>)/R<sub>θJA</sub>. Operating at the absolute maximum T<sub>J</sub> of 150°C can affect reliability.
- (3) The package thermal impedance is calculated in accordance with JESD 51-7.
- (4) Maximum power dissipation is a function of T<sub>J(max)</sub>, R<sub>θJC</sub>, and T<sub>C</sub>. The maximum allowable power dissipation at any allowable ambient temperature is P<sub>D</sub> = (T<sub>J(max)</sub> – T<sub>C</sub>) / R<sub>θJC</sub>. Operating at the absolute maximum T<sub>J</sub> of 150°C can affect reliability.
- (5) The package thermal impedance is calculated in accordance with MIL-STD-883.

## 6.5 Thermal Information (TL062)

| THERMAL METRIC <sup>(1)</sup> |                                                              | TL062    |          |         |            |           |           | UNIT |
|-------------------------------|--------------------------------------------------------------|----------|----------|---------|------------|-----------|-----------|------|
|                               |                                                              | D (SOIC) | P (PDIP) | PS (SO) | PW (TSSOP) | FK (LCCC) | JG (CDIP) |      |
|                               |                                                              | 8 PINS   | 8 PINS   | 8 PINS  | 8 PINS     | 20 PINS   | 8 PINS    |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance <sup>(2) (3)</sup>    | 97       | 85       | 95      | 149        | —         | —         | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance <sup>(4) (5)</sup> | —        | —        | —       | —          | 5.61      | 14.5      | °C/W |

## 6.6 Thermal Information (TL064)

| THERMAL METRIC <sup>(1)</sup> |                                                              | TL064    |          |         |         |            |           |          |         | UNIT |
|-------------------------------|--------------------------------------------------------------|----------|----------|---------|---------|------------|-----------|----------|---------|------|
|                               |                                                              | D (SOIC) | N (PDIP) | NS (SO) | PS (SO) | PW (TSSOP) | FK (LCCC) | J (CDIP) | W (CFP) |      |
|                               |                                                              | 14 PINS  | 14 PINS  | 14 PINS | 8 PINS  | 14 PINS    | 20 PINS   | 14 PINS  | 14 PINS |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance <sup>(2) (3)</sup>    | 86       | 80       | 76      | 95      | 113        | —         | —        | —       | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance <sup>(2) (3)</sup> | —        | —        | —       | —       | —          | 5.61      | 15.05    | 14.65   | °C/W |

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).
- (2) Maximum power dissipation is a function of T<sub>J(max)</sub>, R<sub>θJC</sub>, and T<sub>C</sub>. The maximum allowable power dissipation at any allowable ambient temperature is P<sub>D</sub> = (T<sub>J(max)</sub> – T<sub>C</sub>) / R<sub>θJC</sub>. Operating at the absolute maximum T<sub>J</sub> of 150°C can affect reliability.
- (3) The package thermal impedance is calculated in accordance with MIL-STD-883.

## 6.7 Electrical Characteristics for TL06xC and TL06xxC

$V_{CC\pm} = \pm 15\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  to  $(V_{CC+} + V_{CC-}) / 2$  (unless otherwise noted)

| PARAMETER                                                                     | TEST CONDITIONS <sup>(1)</sup>                                                                             | TL061C, TL062C, TL064C    |            |     | TL061AC, TL062AC, TL064AC |            |     | UNIT                         |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------|------------|-----|---------------------------|------------|-----|------------------------------|
|                                                                               |                                                                                                            | MIN                       | TYP        | MAX | MIN                       | TYP        | MAX |                              |
| $V_{IO}$ Input offset voltage                                                 | $V_O = 0$ , $R_S = 50\ \Omega$                                                                             | $T_A = 25^\circ\text{C}$  |            | 3   | 15                        | 3          | 6   | mV                           |
|                                                                               |                                                                                                            | $T_A = \text{Full range}$ |            |     | 20                        |            | 7.5 |                              |
| $\alpha_{VIO}$ Temperature coefficient of input offset voltage                | $V_O = 0$ , $R_S = 50\ \Omega$ , $T_A = \text{Full range}$                                                 | 10                        |            |     | 10                        |            |     | $\mu\text{V}/^\circ\text{C}$ |
| $I_{IO}$ <sup>(3)</sup> Input offset current                                  | $V_O = 0$                                                                                                  | $T_A = 25^\circ\text{C}$  |            | 5   | 200                       | 5          | 100 | pA                           |
|                                                                               |                                                                                                            | $T_A = \text{Full range}$ |            |     | 5                         |            | 3   | nA                           |
| $I_{IB}$ <sup>(3)</sup> Input bias current <sup>(2)</sup>                     | $V_O = 0$                                                                                                  | $T_A = 25^\circ\text{C}$  |            | 30  | 400                       | 30         | 200 | pA                           |
|                                                                               |                                                                                                            | $T_A = \text{Full range}$ |            |     | 10                        |            | 7   | nA                           |
| $V_{ICR}$ Common-mode input voltage range                                     | $T_A = 25^\circ\text{C}$                                                                                   | $\pm 11$                  | –12 to 15  |     | $\pm 11$                  | –12 to 15  |     | V                            |
| $V_{OM}$ Maximum peak output voltage swing                                    | $R_L = 10\text{ k}\Omega$ , $T_A = 25^\circ\text{C}$                                                       | $\pm 10$                  | $\pm 13.5$ |     | $\pm 10$                  | $\pm 13.5$ |     | V                            |
|                                                                               | $R_L \geq 10\text{ k}\Omega$ , $T_A = \text{Full range}$                                                   | $\pm 10$                  |            |     | $\pm 10$                  |            |     |                              |
| $A_{VD}$ Large-signal differential voltage amplification                      | $V_O = \pm 10\text{ V}$ , $R_L \geq 2\text{ k}\Omega$                                                      | $T_A = 25^\circ\text{C}$  |            | 3   | 6                         | 4          | 6   | V/mV                         |
|                                                                               |                                                                                                            | $T_A = \text{Full range}$ |            | 3   |                           | 4          |     |                              |
| $B_1$ Unity-gain bandwidth                                                    | $R_L = 10\text{ k}\Omega$ , $T_A = 25^\circ\text{C}$                                                       | 1                         |            |     | 1                         |            |     | MHz                          |
| $r_i$ Input resistance                                                        | $T_A = 25^\circ\text{C}$                                                                                   | $10^{12}$                 |            |     | $10^{12}$                 |            |     | $\Omega$                     |
| CMRR Common-mode rejection ratio                                              | $V_{IC} = V_{ICRmin}$ , $V_O = 0$ , $R_S = 50\ \Omega$ , $T_A = 25^\circ\text{C}$                          | 70                        | 86         |     | 80                        | 86         |     | dB                           |
| $k_{SVR}$ Supply-voltage rejection ratio ( $\Delta V_{CC\pm}/\Delta V_{IO}$ ) | $V_{CC} = \pm 9\text{ V}$ to $\pm 15\text{ V}$ , $V_O = 0$ , $R_S = 50\ \Omega$ , $T_A = 25^\circ\text{C}$ | 70                        | 95         |     | 80                        | 95         |     | dB                           |
| $P_D$ Total power dissipation (each amplifier)                                | $V_O = 0$ , No load, $T_A = 25^\circ\text{C}$                                                              | 6                         |            |     | 7.5                       | 6          | 7.5 | mW                           |
| $I_{CC}$ Supply current (each amplifier)                                      | $V_O = 0$ , No load, $T_A = 25^\circ\text{C}$                                                              | 200                       |            |     | 250                       | 200        | 250 | $\mu\text{A}$                |
| $V_{O1}/V_{O2}$ Crosstalk attenuation                                         | $A_{VD} = 100$ , $T_A = 25^\circ\text{C}$                                                                  | 120                       |            |     | 120                       |            |     | dB                           |

- (1) All characteristics are measured under open-loop conditions with zero common-mode input voltage unless otherwise specified. Full range for  $T_A$  is  $0^\circ\text{C}$  to  $70^\circ\text{C}$  for TL06xC, TL06xAC, and TL06xBC and  $-40^\circ\text{C}$  to  $85^\circ\text{C}$  for TL06xI.
- (2) Input bias currents of an FET-input operational amplifier are normal junction reverse currents, which are temperature sensitive, as shown in Figure 6-12. Pulse techniques are used to maintain the junction temperature as close to the ambient temperature as possible.
- (3) Specified by design and characterization; not production tested.



## 6.8 Electrical Characteristics for TL06xxC and TL06xI

$V_{CC\pm} = \pm 15\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  to  $(V_{CC+} + V_{CC-}) / 2$  (unless otherwise noted)

| PARAMETER                        |                                                                       | TEST CONDITIONS <sup>(1)</sup>                                                                           |                             | TL061BC, TL062BC, TL064BC |           |     | TL061I, TL062I, TL064I |           |       | UNIT |
|----------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------|-----------|-----|------------------------|-----------|-------|------|
|                                  |                                                                       |                                                                                                          |                             | MIN                       | TYP       | MAX | MIN                    | TYP       | MAX   |      |
| V <sub>IO</sub>                  | Input offset voltage                                                  | V <sub>O</sub> = 0, R <sub>S</sub> = 50 Ω                                                                | T <sub>A</sub> = 25°C       | 2                         | 3         |     | 3                      | 6         | mV    |      |
|                                  |                                                                       |                                                                                                          | T <sub>A</sub> = Full range |                           | 5         |     |                        | 9         |       |      |
| α <sub>VIO</sub>                 | Temperature coefficient of input offset voltage                       | V <sub>O</sub> = 0, R <sub>S</sub> = 50 Ω, T <sub>A</sub> = Full range                                   |                             | 10                        |           |     | 10                     |           | μV/°C |      |
| I <sub>IO</sub> <sup>(3)</sup>   | Input offset current                                                  | V <sub>O</sub> = 0                                                                                       | T <sub>A</sub> = 25°C       | 5                         | 100       |     | 5                      | 100       | pA    |      |
|                                  |                                                                       |                                                                                                          | T <sub>A</sub> = Full range | 3                         |           |     | 10                     |           | nA    |      |
| I <sub>IB</sub> <sup>(3)</sup>   | Input bias current <sup>(2)</sup>                                     | V <sub>O</sub> = 0                                                                                       | T <sub>A</sub> = 25°C       | 30                        | 200       |     | 30                     | 200       | pA    |      |
|                                  |                                                                       |                                                                                                          | T <sub>A</sub> = Full range | 7                         |           |     | 20                     |           | nA    |      |
| V <sub>ICR</sub>                 | Common-mode input voltage range                                       | T <sub>A</sub> = 25°C                                                                                    |                             | ±11                       | −12 to 15 |     | ±11                    | −12 to 15 | V     |      |
| V <sub>OM</sub>                  | Maximum peak output voltage swing                                     | R <sub>L</sub> = 10 kΩ, T <sub>A</sub> = 25°C                                                            |                             | ±10                       | ±13.5     |     | ±10                    | ±13.5     | V     |      |
|                                  |                                                                       | R <sub>L</sub> ≥ 10 kΩ, T <sub>A</sub> = Full range                                                      |                             | ±10                       |           |     | ±10                    |           |       |      |
| A <sub>VD</sub>                  | Large-signal differential voltage amplification                       | V <sub>O</sub> = ±10 V, R <sub>L</sub> ≥ 2 kΩ                                                            | T <sub>A</sub> = 25°C       | 4                         | 6         |     | 4                      | 6         | V/mV  |      |
|                                  |                                                                       |                                                                                                          | T <sub>A</sub> = Full range | 4                         |           |     | 4                      |           |       |      |
| B <sub>1</sub>                   | Unity-gain bandwidth                                                  | R <sub>L</sub> = 10 kΩ, T <sub>A</sub> = 25°C                                                            |                             | 1                         |           |     | 1                      |           | MHz   |      |
| r <sub>i</sub>                   | Input resistance                                                      | T <sub>A</sub> = 25°C                                                                                    |                             | 10 <sup>12</sup>          |           |     | 10 <sup>12</sup>       |           | Ω     |      |
| CMRR                             | Common-mode rejection ratio                                           | V <sub>IC</sub> = V <sub>ICRmin</sub> , V <sub>O</sub> = 0, R <sub>S</sub> = 50 Ω, T <sub>A</sub> = 25°C |                             | 80                        | 86        |     | 80                     | 86        | dB    |      |
| k <sub>SVR</sub>                 | Supply-voltage rejection ratio (ΔV <sub>CC±</sub> /ΔV <sub>IO</sub> ) | V <sub>CC</sub> = ±9 V to ±15 V, V <sub>O</sub> = 0, R <sub>S</sub> = 50 Ω, T <sub>A</sub> = 25°C        |                             | 80                        | 95        |     | 80                     | 95        | dB    |      |
| P <sub>D</sub>                   | Total power dissipation (each amplifier)                              | V <sub>O</sub> = 0, No load, T <sub>A</sub> = 25°C                                                       |                             | 6                         | 7.5       |     | 6                      | 7.5       | mW    |      |
| I <sub>CC</sub>                  | Supply current (each amplifier)                                       | V <sub>O</sub> = 0, No load, T <sub>A</sub> = 25°C                                                       |                             | 200                       |           | 250 | 200                    |           | 250   | μA   |
| V <sub>O1</sub> /V <sub>O2</sub> | Crosstalk attenuation                                                 | A <sub>VD</sub> = 100, T <sub>A</sub> = 25°C                                                             |                             | 120                       |           |     | 120                    |           |       | dB   |

- (1) All characteristics are measured under open-loop conditions with zero common-mode input voltage, unless otherwise specified. Full range for  $T_A$  is  $0^\circ\text{C}$  to  $70^\circ\text{C}$  for TL06xC, TL06xAC, and TL06xBC and  $-40^\circ\text{C}$  to  $85^\circ\text{C}$  for TL06xI.
- (2) Input bias currents of an FET-input operational amplifier are normal junction reverse currents, which are temperature sensitive, as shown in Figure 6-12. Pulse techniques are used to maintain the junction temperature as close to the ambient temperature as possible.
- (3) Assured by design and characterization; not production tested.

## 6.9 Electrical Characteristics for TL06xM

$V_{CC\pm} = \pm 15\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  to  $(V_{CC+} + V_{CC-}) / 2$  (unless otherwise noted)

| PARAMETER                        |                                                                       | TEST CONDITIONS <sup>(2)</sup>                                                                           |                                 | TL061M, TL062M    |           |                   | TL064M           |           |     | UNIT  |
|----------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------|-------------------|-----------|-------------------|------------------|-----------|-----|-------|
|                                  |                                                                       |                                                                                                          |                                 | MIN               | TYP       | MAX               | MIN              | TYP       | MAX |       |
| V <sub>IO</sub>                  | Input offset voltage                                                  | V <sub>O</sub> = 0, R <sub>S</sub> = 50 Ω                                                                | T <sub>A</sub> = 25°C           | 3                 |           | 6                 | 3                |           | 9   | mV    |
|                                  |                                                                       |                                                                                                          | T <sub>A</sub> = −55°C to 125°C |                   |           | 9                 |                  |           | 15  |       |
| α <sub>VIO</sub>                 | Temperature coefficient of input offset voltage                       | V <sub>O</sub> = 0, R <sub>S</sub> = 50 Ω, T <sub>A</sub> = −55°C to 125°C                               |                                 | 10                |           |                   | 10               |           |     | μV/°C |
| I <sub>IO</sub> <sup>(4)</sup>   | Input offset current                                                  | V <sub>O</sub> = 0                                                                                       | T <sub>A</sub> = 25°C           | 5                 |           | 100               | 5                |           | 100 | pA    |
|                                  |                                                                       |                                                                                                          | T <sub>A</sub> = −55°C          | 20 <sup>(1)</sup> |           | 20 <sup>(1)</sup> |                  | nA        |     |       |
|                                  |                                                                       |                                                                                                          | T <sub>A</sub> = 125°C          | 20                |           | 20                |                  |           |     |       |
| I <sub>IB</sub> <sup>(4)</sup>   | Input bias current <sup>(3)</sup>                                     | V <sub>O</sub> = 0                                                                                       | T <sub>A</sub> = 25°C           | 30                |           | 200               | 30               |           | 200 | pA    |
|                                  |                                                                       |                                                                                                          | T <sub>A</sub> = −55°C          | 50 <sup>(1)</sup> |           | 50 <sup>(1)</sup> |                  | nA        |     |       |
|                                  |                                                                       |                                                                                                          | T <sub>A</sub> = 125°C          | 50                |           | 50                |                  |           |     |       |
| V <sub>ICR</sub>                 | Common-mode input voltage range                                       | T <sub>A</sub> = 25°C                                                                                    |                                 | ±11               | −12 to 15 |                   | ±11              | −12 to 15 |     | V     |
| V <sub>OM</sub>                  | Maximum peak output voltage swing                                     | R <sub>L</sub> = 10 kΩ, T <sub>A</sub> = 25°C                                                            |                                 | ±10               | ±13.5     |                   | ±10              | ±13.5     |     | V     |
|                                  |                                                                       | R <sub>L</sub> ≥ 10 kΩ, T <sub>A</sub> = −55°C to 125°C                                                  |                                 | ±10               |           |                   | ±10              |           |     |       |
| A <sub>VD</sub>                  | Large-signal differential voltage amplification                       | V <sub>O</sub> = ±10 V, R <sub>L</sub> ≥ 2 kΩ                                                            | T <sub>A</sub> = 25°C           | 4                 | 6         |                   | 4                | 6         |     | V/mV  |
|                                  |                                                                       |                                                                                                          | T <sub>A</sub> = −55°C to 125°C | 4                 |           |                   | 4                |           |     |       |
| B <sub>1</sub>                   | Unity-gain bandwidth                                                  | R <sub>L</sub> = 10 kΩ, T <sub>A</sub> = 25°C                                                            |                                 | 1                 |           |                   | 1                |           |     | MHz   |
| r <sub>i</sub>                   | Input resistance                                                      | T <sub>A</sub> = 25°C                                                                                    |                                 | 10 <sup>12</sup>  |           |                   | 10 <sup>12</sup> |           |     | Ω     |
| CMRR                             | Common-mode rejection ratio                                           | V <sub>IC</sub> = V <sub>ICRmin</sub> , V <sub>O</sub> = 0, R <sub>S</sub> = 50 Ω, T <sub>A</sub> = 25°C |                                 | 80                | 86        |                   | 80               | 86        |     | dB    |
| k <sub>SVR</sub>                 | Supply-voltage rejection ratio (ΔV <sub>CC±</sub> /ΔV <sub>IO</sub> ) | V <sub>CC</sub> = ±9 V to ±15 V, V <sub>O</sub> = 0, R <sub>S</sub> = 50 Ω, T <sub>A</sub> = 25°C        |                                 | 80                | 95        |                   | 80               | 95        |     | dB    |
| P <sub>D</sub>                   | Total power dissipation (each amplifier)                              | V <sub>O</sub> = 0, No load, T <sub>A</sub> = 25°C                                                       |                                 | 6                 |           |                   | 7.5              | 6         | 7.5 | mW    |
| I <sub>CC</sub>                  | Supply current (each amplifier)                                       | V <sub>O</sub> = 0, No load, T <sub>A</sub> = 25°C                                                       |                                 | 200               |           |                   | 250              | 200       | 250 | μA    |
| V <sub>O1</sub> /V <sub>O2</sub> | Crosstalk attenuation                                                 | A <sub>VD</sub> = 100, T <sub>A</sub> = 25°C                                                             |                                 | 120               |           |                   |                  | 120       |     | dB    |

- (1) This parameter is not production tested.  
 (2) All characteristics are measured under open-loop conditions, with zero common-mode voltage, unless otherwise specified.  
 (3) Input bias currents of an FET-input operational amplifier are normal junction reverse currents, which are temperature sensitive, as shown in Figure 6-12. Pulse techniques are used to maintain the junction temperature as close to the ambient temperature as possible.  
 (4) Specified by design and characterization; not production tested.

## 6.10 Operating Characteristics

$V_{CC\pm} = \pm 15\text{ V}$ ,  $T_A = 25^\circ\text{C}$ ,  $R_L = 10\text{ k}\Omega$  to  $(V_{CC+} + V_{CC-}) / 2$

| PARAMETER                                 | TEST CONDITIONS                                                                          | MIN | TYP | MAX | UNIT                   |
|-------------------------------------------|------------------------------------------------------------------------------------------|-----|-----|-----|------------------------|
| SR Slew rate at unity gain <sup>(1)</sup> | $V_I = 10\text{ V}$ , $R_L = 10\text{ k}\Omega$ , $C_L = 100\text{ pF}$ , see Figure 7-1 | 1.5 | 3.5 |     | V/ $\mu\text{s}$       |
| $t_r$ Rise-time                           | $V_I = 20\text{ V}$ , $R_L = 10\text{ k}\Omega$ , $C_L = 100\text{ pF}$ , see Figure 7-1 |     | 0.2 |     | $\mu\text{s}$          |
| Overshoot factor                          |                                                                                          |     | 10% |     |                        |
| $V_n$ Equivalent input noise voltage      | $R_S = 20\text{ }\Omega$ , $f = 1\text{ kHz}$                                            |     | 30  |     | nV/ $\sqrt{\text{Hz}}$ |

- (1) Slew rate at  $-55^\circ\text{C}$  to  $125^\circ\text{C}$  is  $0.7\text{ V}/\mu\text{s}$  min.

## Typical Characteristics

Data at high and low temperatures are applicable only within the specified operating free-air temperature ranges of the various devices.

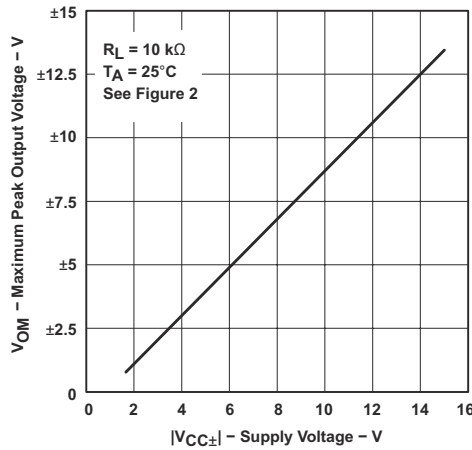


Figure 6-1. Maximum Peak Output Voltage vs Supply Voltage

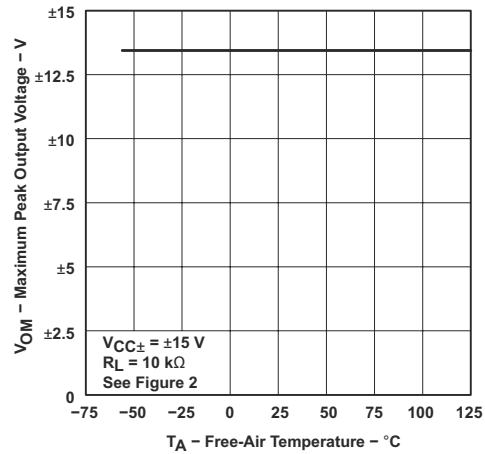


Figure 6-2. Maximum Peak Output Voltage vs Free-Air Temperature

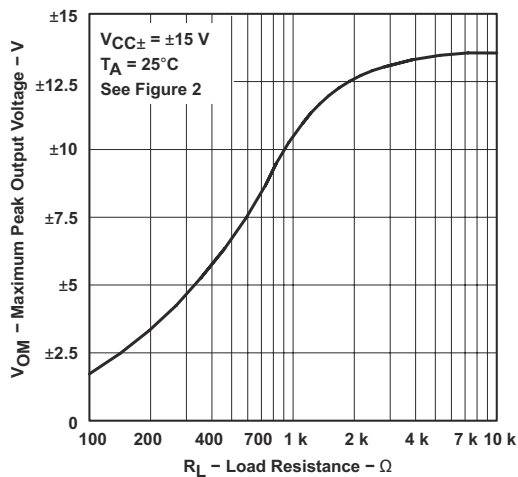


Figure 6-3. Maximum Peak Output Voltage vs Load Resistance

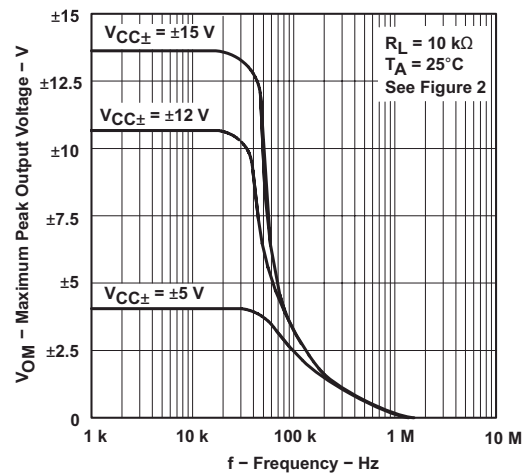


Figure 6-4. Maximum Peak Output Voltage vs Frequency

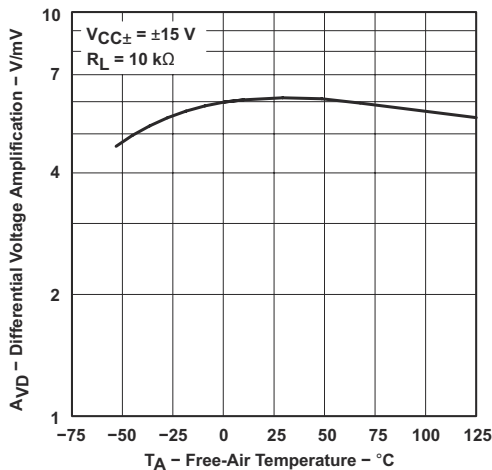


Figure 6-5. Differential Voltage Amplification vs Free-Air Temperature

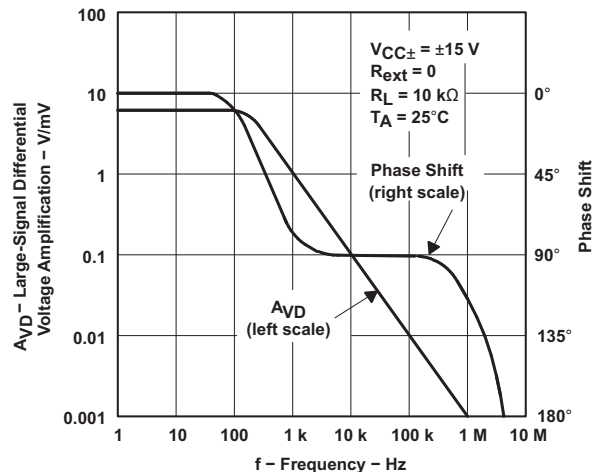


Figure 6-6. Large-Signal Differential Voltage Amplification and Phase Shift vs Frequency

## Typical Characteristics (continued)

Data at high and low temperatures are applicable only within the specified operating free-air temperature ranges of the various devices.

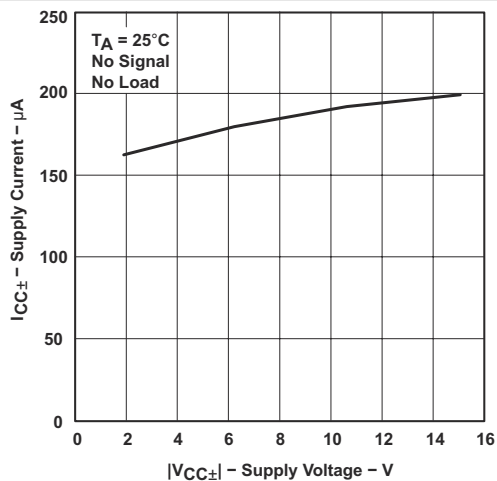


Figure 6-7. Supply Current vs Supply Voltage

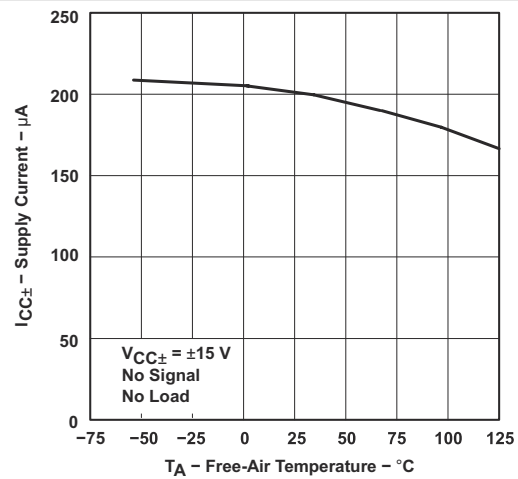


Figure 6-8. Supply Current vs Free-Air Temperature

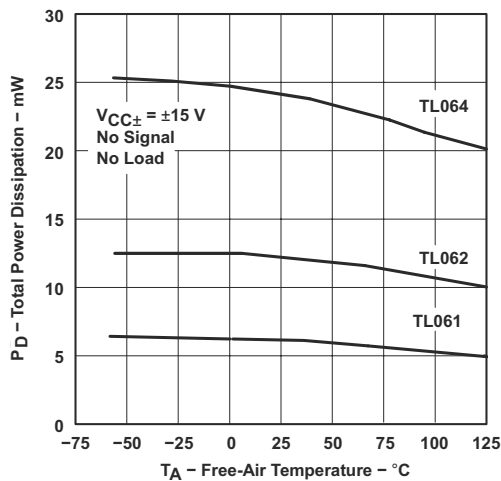


Figure 6-9. Total Power Dissipation vs Free-Air Temperature

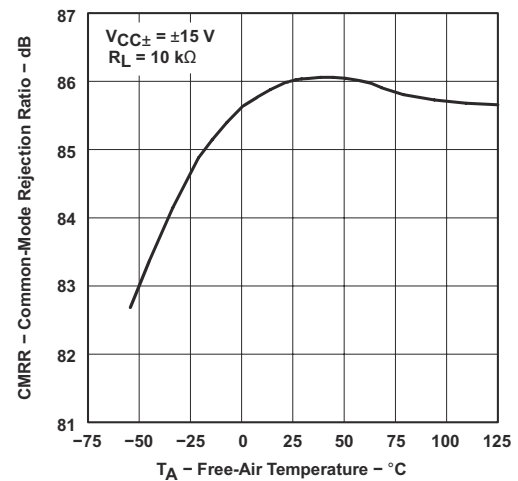


Figure 6-10. All Except TL06\_C Common-Mode Rejection Ratio vs Free-Air Temperature

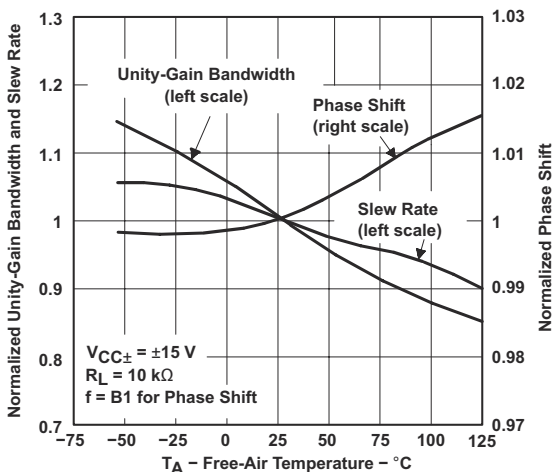


Figure 6-11. Normalized Unity-Gain Bandwidth, Slew Rate, and Phase Shift vs Free-Air Temperature

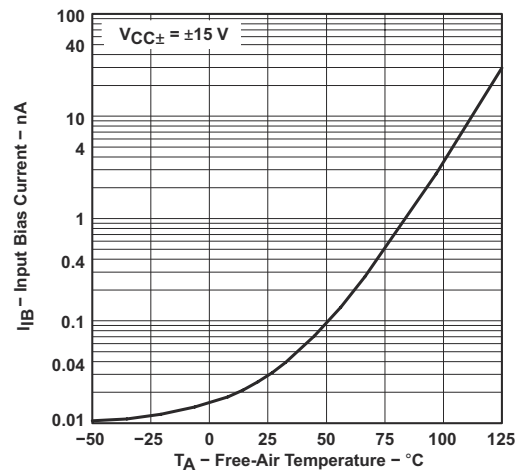


Figure 6-12. Input Bias Current vs Free-Air Temperature

## Typical Characteristics (continued)

Data at high and low temperatures are applicable only within the specified operating free-air temperature ranges of the various devices.

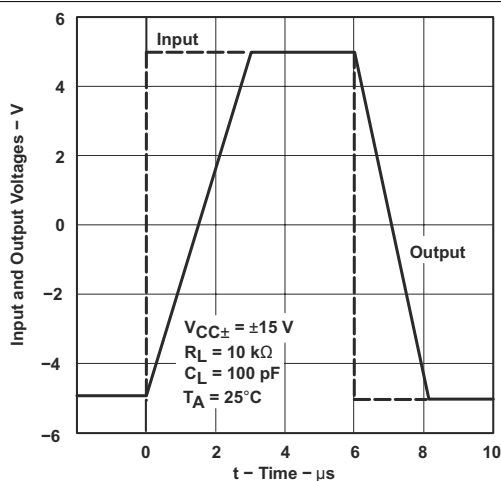


Figure 6-13. Voltage-Follower Large-Signal Pulse Response vs Time

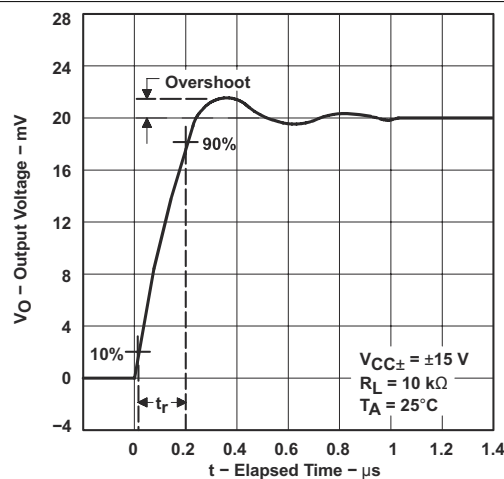


Figure 6-14. Output Voltage vs Elapsed Time

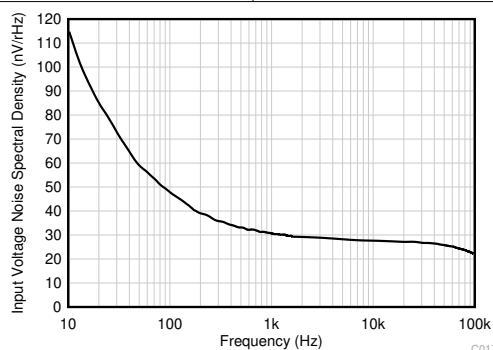


Figure 6-15. Equivalent Input Noise Voltage vs Frequency

## 7 Parameter Measurement Information

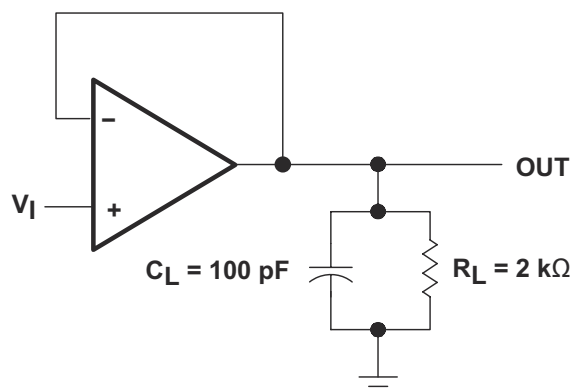


Figure 7-1. Unity-Gain Amplifier

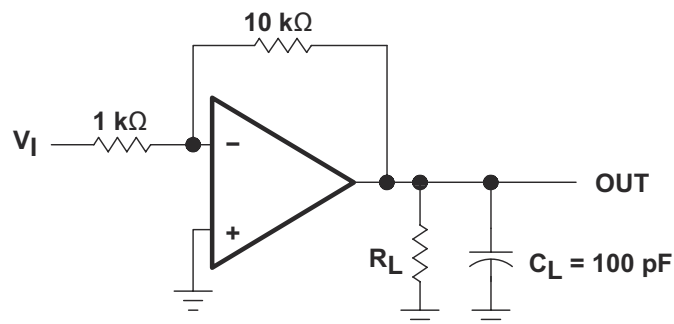


Figure 7-2. Gain-of-10 Inverting Amplifier

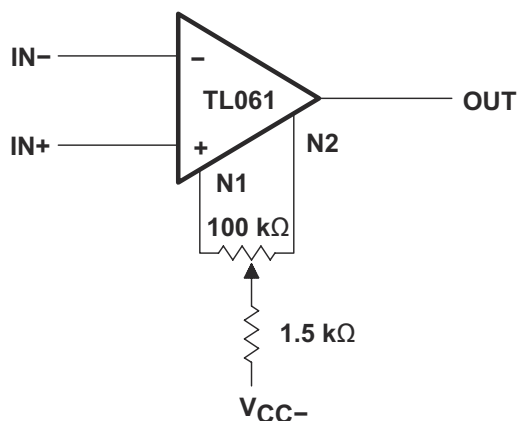


Figure 7-3. Input Offset-Voltage Null Circuit

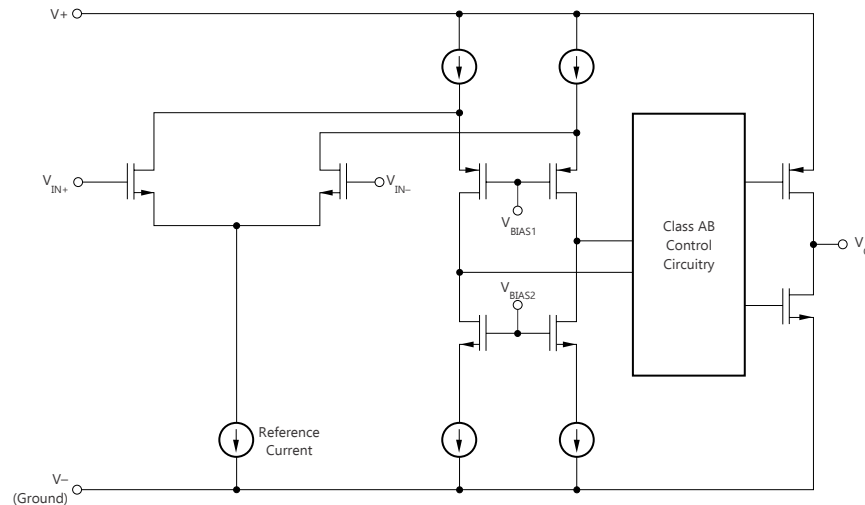
## 8 Detailed Description

### 8.1 Overview

The TL06x (TL061, TL062, and TL064) family of industry-standard operational amplifiers (op amps) mirror the TL07x and TL08x family of op amps with lower power consumption. These devices provide outstanding value for cost-sensitive applications, featuring high input impedance, wide bandwidth, high slew rate, and low input offset and input bias currents. High ESD (1.5 kV, HBM), integrated EMI and RF filters, and wide temperature operation enable the TL06x devices to be used in rugged and environmentally-demanding applications.

The C-suffix devices are characterized for operation from 0°C to 70°C. The I-suffix devices are characterized for operation from –40°C to 85°C, and the M-suffix devices are characterized for operation over the full military temperature range of –55°C to 125°C.

### 8.2 Functional Block Diagram



### 8.3 Feature Description

#### 8.3.1 Common-Mode Rejection Ratio

The common-mode rejection ratio (CMRR) of an amplifier is a measure of how well the device rejects unwanted input signals common to both input leads. It is found by taking the ratio of the change in input offset voltage to the change in the input voltage and converting to decibels. Ideally the CMRR is infinite, but in practice, amplifiers are designed to have it as high as possible. The CMRR of this device is 86 dB.

#### 8.3.2 Slew Rate

The slew rate is the rate at which an operational amplifier can change its output when there is a change on the input. These devices have a 3.5-V/μs slew rate.

### 8.4 Device Functional Modes

These devices are powered on when the supply is connected. This device can be operated as a single supply operational amplifier or dual supply amplifier depending on the application.

## 9 Applications and Implementation

### Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

### 9.1 Application Information

The TL06x series of operational amplifiers can be used in countless applications. The few applications in this section show principles used in all applications of these parts.

### 9.2 Typical Applications

#### 9.2.1 Inverting Amplifier Application

A typical application for an operational amplifier in an inverting amplifier. This amplifier takes a positive voltage on the input, and makes it a negative voltage of the same magnitude. In the same manner, it also makes negative voltages positive.

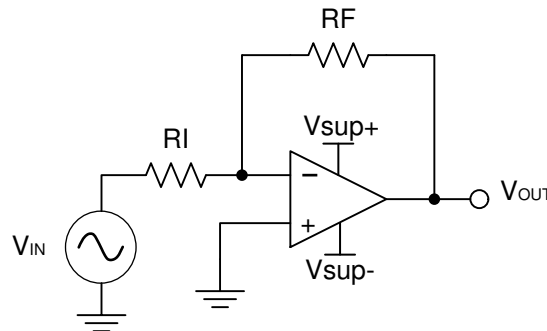


Figure 9-1. Schematic for Inverting Amplifier Application

##### 9.2.1.1 Design Requirements

The supply voltage must be chosen such that it is larger than the input voltage range and output range. For instance, this application will scale a signal of  $\pm 0.5$  V to  $\pm 1.8$  V. Setting the supply at  $\pm 12$  V is sufficient to accommodate this application.

##### 9.2.1.2 Detailed Design Procedure

Determine the gain required by the inverting amplifier:

$$A_V = \frac{V_{OUT}}{V_{IN}} \quad (1)$$

$$A_V = \frac{1.8}{-0.5} = -3.6 \quad (2)$$

Once the desired gain is determined, choose a value for  $R_I$  or  $R_F$ . Choose a value in the k $\Omega$  range to limit currents in the amplifier circuit to the mA range. This example will choose 10 k $\Omega$  for  $R_I$  which means 36 k $\Omega$  will be used for  $R_F$ . This was determined by Equation 3.

$$A_V = -\frac{R_F}{R_I} \quad (3)$$



### 9.2.1.3 Application Curve

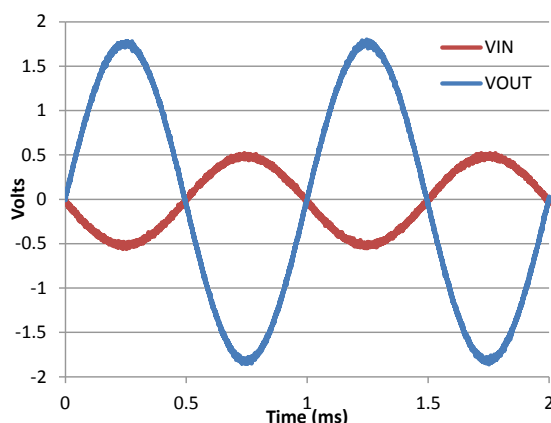


Figure 9-2. Input and Output Voltages of the Inverting Amplifier

## 9.3 System Examples

### 9.3.1 General Applications

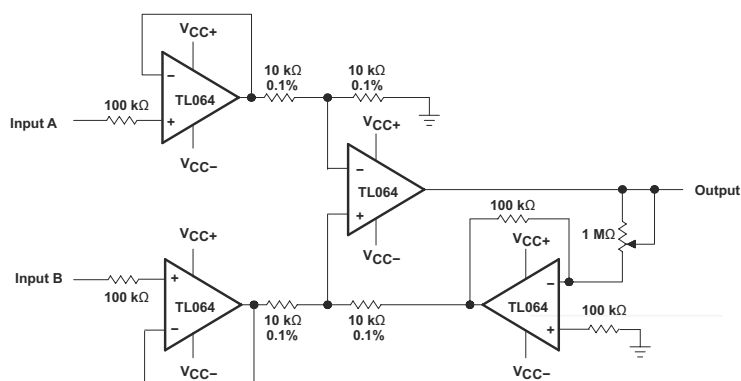


Figure 9-3. Instrumentation Amplifier

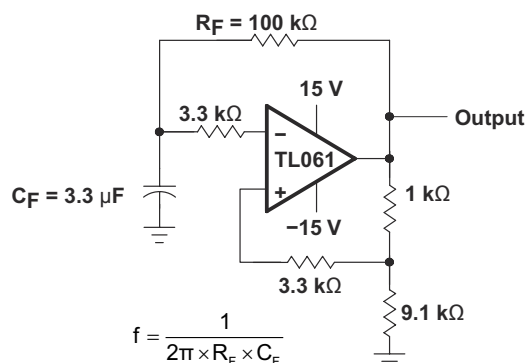


Figure 9-4. 0.5-Hz Square-Wave Oscillator

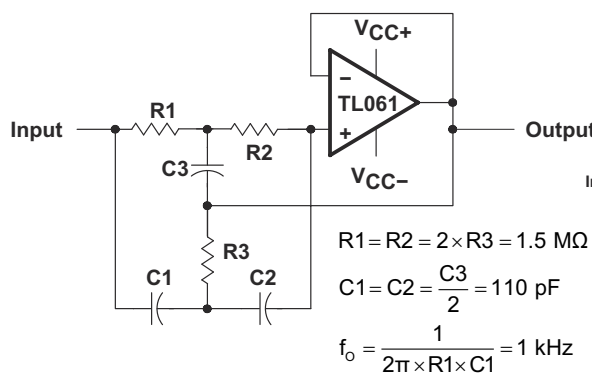


Figure 9-5. High-Q Notch Filter

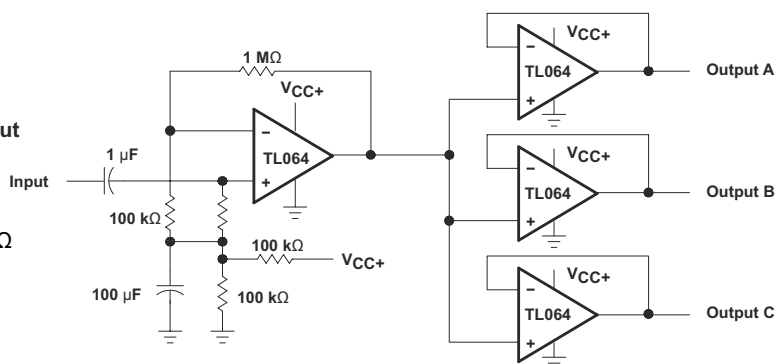


Figure 9-6. Audio-Distribution Amplifier

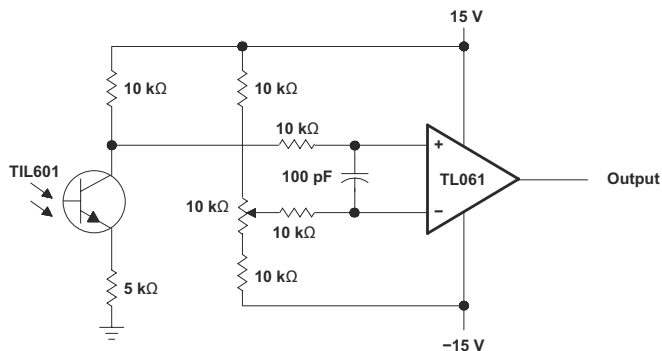


Figure 9-7. Low-Level Light Detector Preamplifier

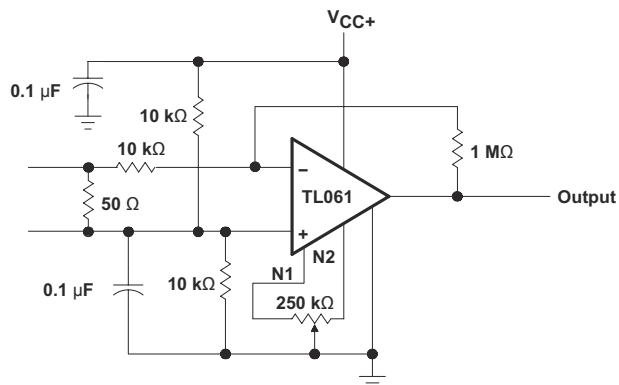


Figure 9-8. AC Amplifier

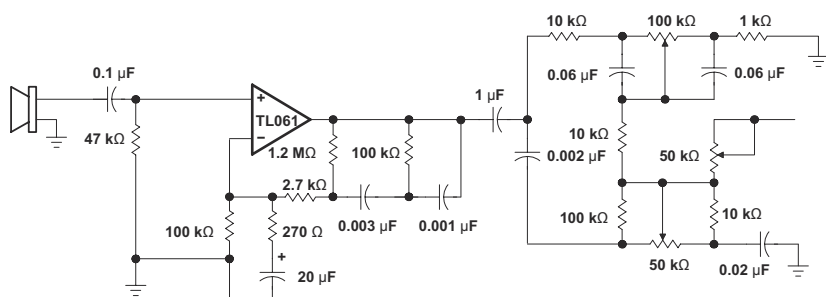


Figure 9-9. Microphone Preamplifier With Tone Control

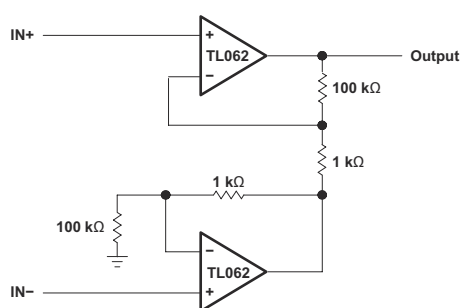


Figure 9-10. Instrumentation Amplifier

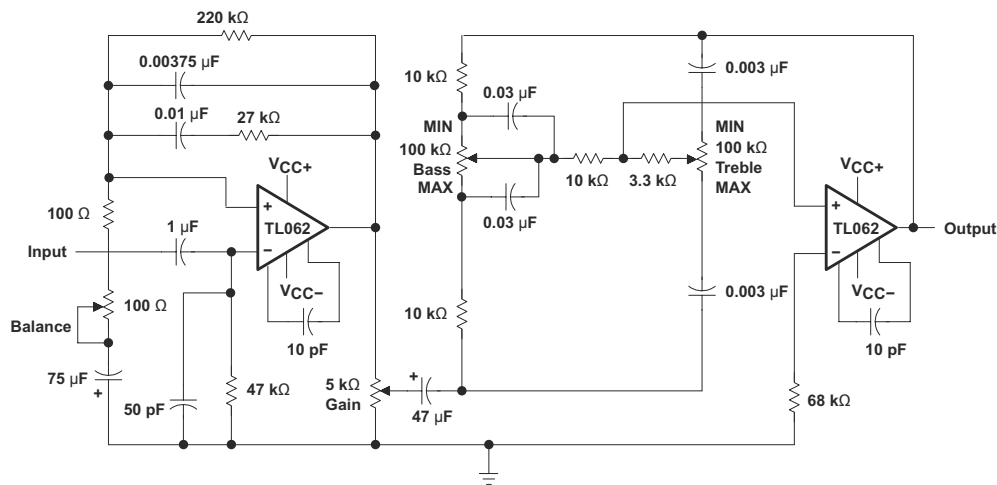
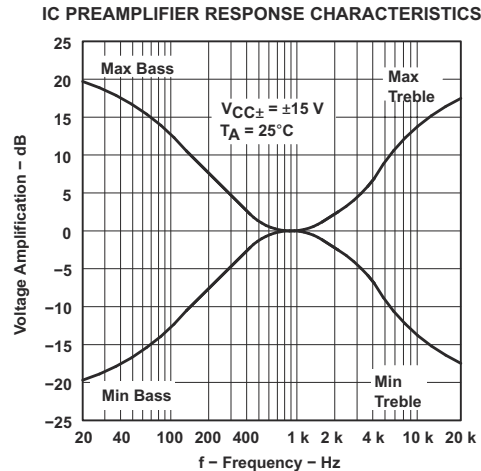


Figure 9-11. IC Preamplifier

## 9.4 Power Supply Recommendations

### CAUTION

Supply voltages larger than 36 V for a single supply, or outside the range of  $\pm 18 \text{ V}$  for a dual supply can permanently damage the device (see the [Absolute Maximum Ratings](#)).

Place 0.1- $\mu\text{F}$  bypass capacitors close to the power-supply pins to reduce errors coupling in from noisy or high impedance power supplies. For more detailed information on bypass capacitor placement, refer to the [Layout](#) section.

## 9.5 Layout

### 9.5.1 Layout Guidelines

For best operational performance of the device, use good PCB layout practices, including:

- Noise can propagate into analog circuitry through the power pins of the circuit as a whole, as well as the operational amplifier. Bypass capacitors are used to reduce the coupled noise by providing low impedance power sources local to the analog circuitry.
  - Connect low-ESR, 0.1- $\mu$ F ceramic bypass capacitors between each supply pin and ground, placed as close to the device as possible. A single bypass capacitor from V+ to ground is applicable for single supply applications.
- Separate grounding for analog and digital portions of circuitry is one of the simplest and most-effective methods of noise suppression. One or more layers on multilayer PCBs are usually devoted to ground planes. A ground plane helps distribute heat and reduces EMI noise pickup. Make sure to physically separate digital and analog grounds, paying attention to the flow of the ground current. For more detailed information, refer to [Circuit Board Layout Techniques](#).
- To reduce parasitic coupling, run the input traces as far away from the supply or output traces as possible. If it is not possible to keep them separate, it is much better to cross the sensitive trace perpendicular as opposed to in parallel with the noisy trace.
- Place the external components as close to the device as possible. Keeping RF and RG close to the inverting input minimizes parasitic capacitance, as shown in [Layout Examples](#).
- Keep the length of input traces as short as possible. Always remember that the input traces are the most sensitive part of the circuit.
- Consider a driven, low-impedance guard ring around the critical traces. A guard ring can significantly reduce leakage currents from nearby traces that are at different potentials.

### 9.5.2 Layout Examples

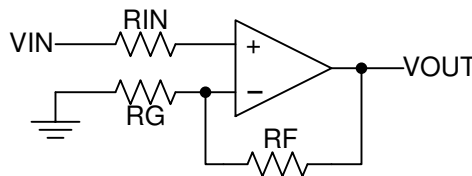


Figure 9-12. Operational Amplifier Schematic for Noninverting Configuration

*A Temet con cuenta si hay q' rediseñar el PCB*

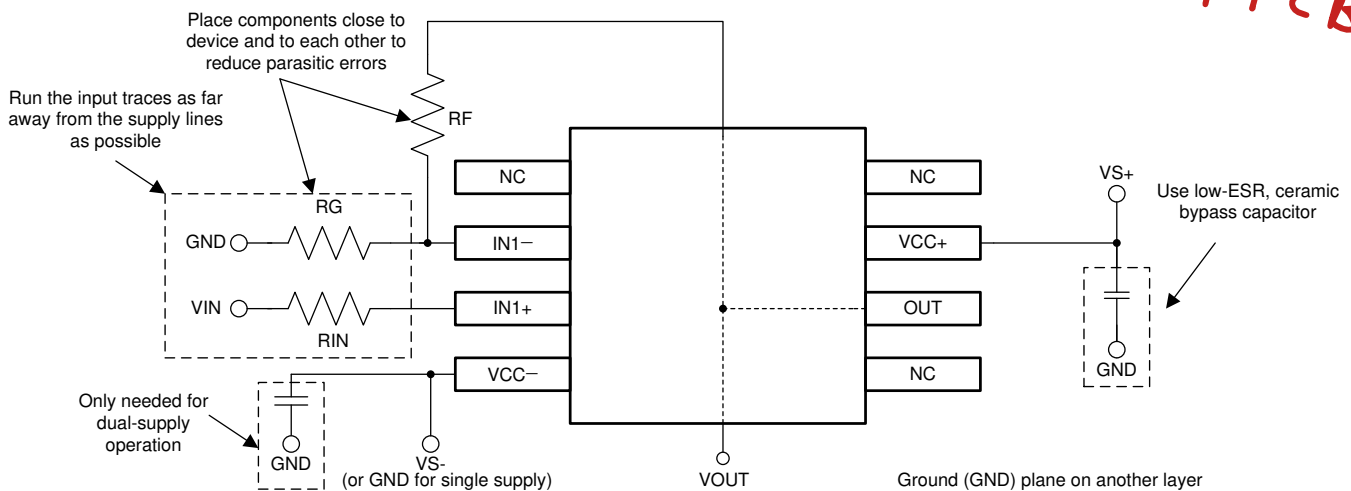


Figure 9-13. Operational Amplifier Board Layout for Noninverting Configuration

## 10 Device and Documentation Support

### 10.1 Documentation Support

#### 10.1.1 Related Documentation

For related documentation, see the following:

- Texas Instruments, [Circuit Board Layout Techniques chapter extracts](#)

### 10.2 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

### 10.3 Trademarks

TI E2E™ is a trademark of Texas Instruments.

All trademarks are the property of their respective owners.

### 10.4 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

### 10.5 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

## 11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser based versions of this data sheet, refer to the left hand navigation.

**PACKAGING INFORMATION**

| Orderable part number     | Status<br>(1) | Material type<br>(2) | Package   Pins | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6)    |
|---------------------------|---------------|----------------------|----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|------------------------|
| <a href="#">81023022A</a> | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023022A<br>TL062MFKB |
| <a href="#">8102302PA</a> | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102302PA<br>TL062M    |
| <a href="#">81023032A</a> | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023032A<br>TL064MFKB |
| <a href="#">8102303CA</a> | Active        | Production           | CDIP (J)   14  | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102303CA<br>TL064MJB  |
| <a href="#">8102303DA</a> | Active        | Production           | CFP (W)   14   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102303DA<br>TL064MWB  |
| <a href="#">TL061ACD</a>  | Obsolete      | Production           | SOIC (D)   8   | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | 061AC                  |
| <a href="#">TL061ACDR</a> | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 061AC                  |
| TL061ACDR.A               | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 061AC                  |
| <a href="#">TL061ACP</a>  | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL061ACP               |
| TL061ACP.A                | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL061ACP               |
| <a href="#">TL061BCP</a>  | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL061BCP               |
| TL061BCP.A                | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL061BCP               |
| TL061BCPE4                | Active        | Production           | PDIP (P)   8   | 50   TUBE             | -           | Call TI                              | Call TI                           | 0 to 70      |                        |
| <a href="#">TL061CD</a>   | Obsolete      | Production           | SOIC (D)   8   | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | TL061C                 |
| <a href="#">TL061CDR</a>  | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL061C                 |
| TL061CDR.A                | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL061C                 |
| <a href="#">TL061CP</a>   | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL061CP                |
| TL061CP.A                 | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL061CP                |
| <a href="#">TL061CPSR</a> | Active        | Production           | SO (PS)   8    | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T061                   |
| TL061CPSR.A               | Active        | Production           | SO (PS)   8    | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T061                   |
| <a href="#">TL061ID</a>   | Obsolete      | Production           | SOIC (D)   8   | -                     | -           | Call TI                              | Call TI                           | -40 to 85    | TL061I                 |
| <a href="#">TL061IDR</a>  | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL061I                 |
| TL061IDR.A                | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL061I                 |
| TL061IDRG4                | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | -           | Call TI                              | Call TI                           | -40 to 85    |                        |
| <a href="#">TL061IP</a>   | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL061IP                |

| Orderable part number       | Status<br>(1) | Material type<br>(2) | Package   Pins | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6) |
|-----------------------------|---------------|----------------------|----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|---------------------|
| TL061IP.A                   | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL061IP             |
| TL061IPE4                   | Active        | Production           | PDIP (P)   8   | 50   TUBE             | -           | Call TI                              | Call TI                           | -40 to 85    |                     |
| <a href="#">TL062ACD</a>    | Obsolete      | Production           | SOIC (D)   8   | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | 062AC               |
| <a href="#">TL062ACDR</a>   | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 062AC               |
| TL062ACDR.A                 | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 062AC               |
| TL062ACDRG4                 | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 062AC               |
| <a href="#">TL062ACP</a>    | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL062ACP            |
| TL062ACP.A                  | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL062ACP            |
| <a href="#">TL062ACPSR</a>  | Active        | Production           | SO (PS)   8    | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T062A               |
| TL062ACPSR.A                | Active        | Production           | SO (PS)   8    | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T062A               |
| <a href="#">TL062BCD</a>    | Obsolete      | Production           | SOIC (D)   8   | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | 062BC               |
| <a href="#">TL062BCDR</a>   | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 062BC               |
| TL062BCDR.A                 | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 062BC               |
| <a href="#">TL062BCP</a>    | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL062BCP            |
| TL062BCP.A                  | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL062BCP            |
| <a href="#">TL062CD</a>     | Obsolete      | Production           | SOIC (D)   8   | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | TL062C              |
| <a href="#">TL062CDR</a>    | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL062C              |
| TL062CDR.A                  | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL062C              |
| TL062CDRE4                  | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL062C              |
| TL062CDRG4                  | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL062C              |
| <a href="#">TL062CP</a>     | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL062CP             |
| TL062CP.A                   | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL062CP             |
| TL062CPE4                   | Active        | Production           | PDIP (P)   8   | 50   TUBE             | -           | Call TI                              | Call TI                           | 0 to 70      |                     |
| <a href="#">TL062CPS</a>    | Active        | Production           | SO (PS)   8    | 80   TUBE             | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T062                |
| TL062CPS.A                  | Active        | Production           | SO (PS)   8    | 80   TUBE             | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T062                |
| <a href="#">TL062CPSR</a>   | Active        | Production           | SO (PS)   8    | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T062                |
| TL062CPSR.A                 | Active        | Production           | SO (PS)   8    | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T062                |
| <a href="#">TL062CPW</a>    | Obsolete      | Production           | TSSOP (PW)   8 | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | T062                |
| <a href="#">TL062CPWR</a>   | Active        | Production           | TSSOP (PW)   8 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T062                |
| TL062CPWR.A                 | Active        | Production           | TSSOP (PW)   8 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T062                |
| <a href="#">TL062CPWRG4</a> | Obsolete      | Production           | TSSOP (PW)   8 | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | T062                |

| Orderable part number     | Status<br>(1) | Material type<br>(2) | Package   Pins | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6)    |
|---------------------------|---------------|----------------------|----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|------------------------|
| <a href="#">TL062ID</a>   | Obsolete      | Production           | SOIC (D)   8   | -                     | -           | Call TI                              | Call TI                           | -40 to 85    | TL062I                 |
| <a href="#">TL062IDR</a>  | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL062I                 |
| TL062IDR.A                | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL062I                 |
| <a href="#">TL062IP</a>   | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL062IP                |
| TL062IP.A                 | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL062IP                |
| <a href="#">TL062IPWR</a> | Active        | Production           | TSSOP (PW)   8 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | Z062                   |
| TL062IPWR.A               | Active        | Production           | TSSOP (PW)   8 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | Z062                   |
| TL062IPWRG4               | Active        | Production           | TSSOP (PW)   8 | 2000   LARGE T&R      | -           | Call TI                              | Call TI                           | -40 to 85    |                        |
| <a href="#">TL062MFKB</a> | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023022A<br>TL062MFKB |
| TL062MFKB.A               | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023022A<br>TL062MFKB |
| <a href="#">TL062MJG</a>  | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | TL062MJG               |
| TL062MJG.A                | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | TL062MJG               |
| <a href="#">TL062MJGB</a> | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102302PA<br>TL062M    |
| TL062MJGB.A               | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102302PA<br>TL062M    |
| <a href="#">TL064ACD</a>  | Obsolete      | Production           | SOIC (D)   14  | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | TL064AC                |
| <a href="#">TL064ACDR</a> | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL064AC                |
| TL064ACDR.A               | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL064AC                |
| <a href="#">TL064ACN</a>  | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL064ACN               |
| TL064ACN.A                | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL064ACN               |
| <a href="#">TL064BCD</a>  | Obsolete      | Production           | SOIC (D)   14  | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | TL064BC                |
| <a href="#">TL064BCDR</a> | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL064BC                |
| TL064BCDR.A               | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL064BC                |
| <a href="#">TL064BCN</a>  | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL064BCN               |
| TL064BCN.A                | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL064BCN               |
| <a href="#">TL064CD</a>   | Obsolete      | Production           | SOIC (D)   14  | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | TL064C                 |
| <a href="#">TL064CDR</a>  | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL064C                 |
| TL064CDR.A                | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL064C                 |
| <a href="#">TL064CN</a>   | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL064CN                |



| Orderable part number       | Status<br>(1) | Material type<br>(2) | Package   Pins  | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6)    |
|-----------------------------|---------------|----------------------|-----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|------------------------|
| TL064CN.A                   | Active        | Production           | PDIP (N)   14   | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL064CN                |
| <a href="#">TL064CNSR</a>   | Active        | Production           | SOP (NS)   14   | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL064                  |
| TL064CNSR.A                 | Active        | Production           | SOP (NS)   14   | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL064                  |
| <a href="#">TL064CPW</a>    | Obsolete      | Production           | TSSOP (PW)   14 | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | T064                   |
| <a href="#">TL064CPWR</a>   | Active        | Production           | TSSOP (PW)   14 | 2000   LARGE T&R      | Yes         | NIPDAU   SN                          | Level-1-260C-UNLIM                | 0 to 70      | T064                   |
| TL064CPWR.A                 | Active        | Production           | TSSOP (PW)   14 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T064                   |
| <a href="#">TL064CPWRG4</a> | Active        | Production           | TSSOP (PW)   14 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | T064                   |
| <a href="#">TL064ID</a>     | Obsolete      | Production           | SOIC (D)   14   | -                     | -           | Call TI                              | Call TI                           | -40 to 85    | TL064I                 |
| <a href="#">TL064IDR</a>    | Active        | Production           | SOIC (D)   14   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL064I                 |
| TL064IDR.A                  | Active        | Production           | SOIC (D)   14   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL064I                 |
| <a href="#">TL064IDRG4</a>  | Obsolete      | Production           | SOIC (D)   14   | -                     | -           | Call TI                              | Call TI                           | -40 to 85    | TL064I                 |
| <a href="#">TL064IN</a>     | Active        | Production           | PDIP (N)   14   | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL064IN                |
| TL064IN.A                   | Active        | Production           | PDIP (N)   14   | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL064IN                |
| TL064INE4                   | Active        | Production           | PDIP (N)   14   | 25   TUBE             | -           | Call TI                              | Call TI                           | -40 to 85    |                        |
| <a href="#">TL064INS</a>    | Active        | Production           | SOP (NS)   14   | 50   TUBE             | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL064I                 |
| TL064INS.A                  | Active        | Production           | SOP (NS)   14   | 50   TUBE             | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL064I                 |
| <a href="#">TL064INSR</a>   | Active        | Production           | SOP (NS)   14   | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL064I                 |
| TL064INSR.A                 | Active        | Production           | SOP (NS)   14   | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL064I                 |
| <a href="#">TL064IPWR</a>   | Active        | Production           | TSSOP (PW)   14 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | Z064                   |
| TL064IPWR.A                 | Active        | Production           | TSSOP (PW)   14 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | Z064                   |
| <a href="#">TL064MFKB</a>   | Active        | Production           | LCCC (FK)   20  | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023032A<br>TL064MFKB |
| TL064MFKB.A                 | Active        | Production           | LCCC (FK)   20  | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023032A<br>TL064MFKB |
| <a href="#">TL064MJ</a>     | Active        | Production           | CDIP (J)   14   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | TL064MJ                |
| TL064MJ.A                   | Active        | Production           | CDIP (J)   14   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | TL064MJ                |
| <a href="#">TL064MJB</a>    | Active        | Production           | CDIP (J)   14   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102303CA<br>TL064MJB  |
| TL064MJB.A                  | Active        | Production           | CDIP (J)   14   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102303CA<br>TL064MJB  |
| <a href="#">TL064MWB</a>    | Active        | Production           | CFP (W)   14    | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102303DA<br>TL064MWB  |

| Orderable part number | Status<br>(1) | Material type<br>(2) | Package   Pins | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6)   |
|-----------------------|---------------|----------------------|----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|-----------------------|
| TL064MWB.A            | Active        | Production           | CFP (W)   14   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102303DA<br>TL064MWB |

<sup>(1)</sup> **Status:** For more details on status, see our [product life cycle](#).

<sup>(2)</sup> **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

<sup>(3)</sup> **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

<sup>(4)</sup> **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

<sup>(5)</sup> **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

<sup>(6)</sup> **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

**Important Information and Disclaimer:**The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

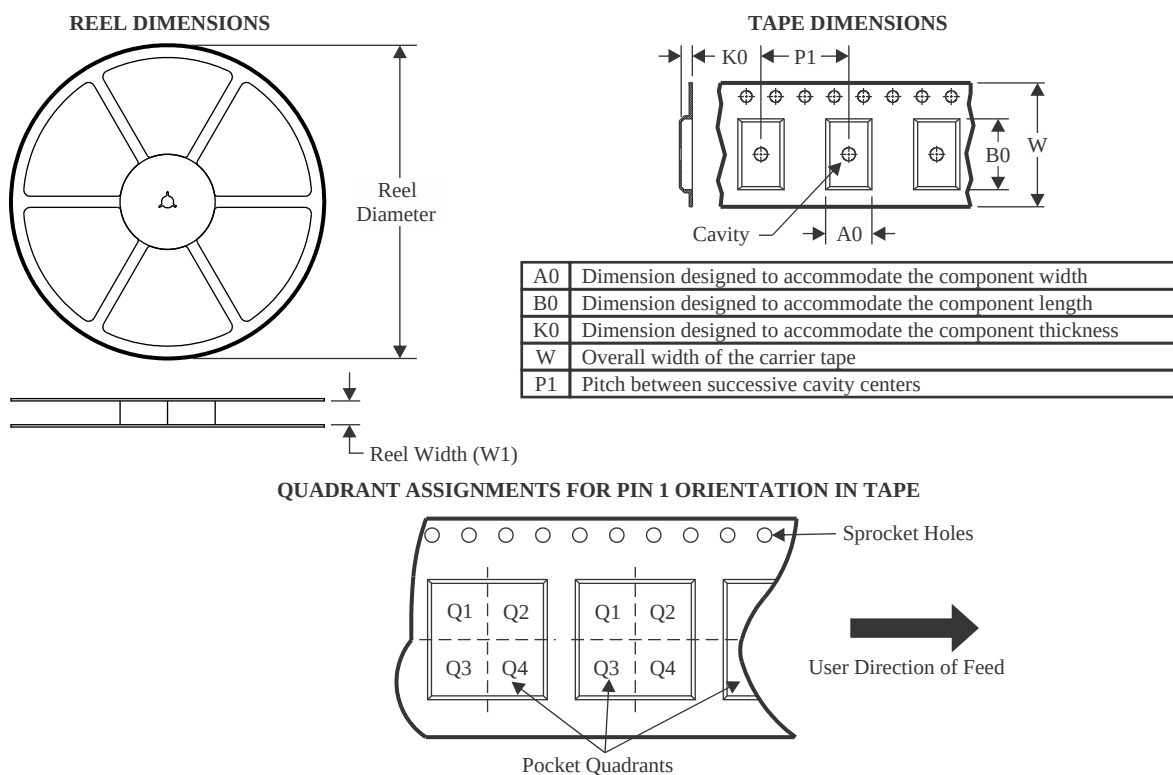
## OTHER QUALIFIED VERSIONS OF TL062, TL062M, TL064, TL064M :

- Catalog : [TL062](#), [TL064](#)
- Military : [TL062M](#), [TL064M](#)

NOTE: Qualified Version Definitions:

- Catalog - TI's standard catalog product
- Military - QML certified for Military and Defense Applications

## TAPE AND REEL INFORMATION

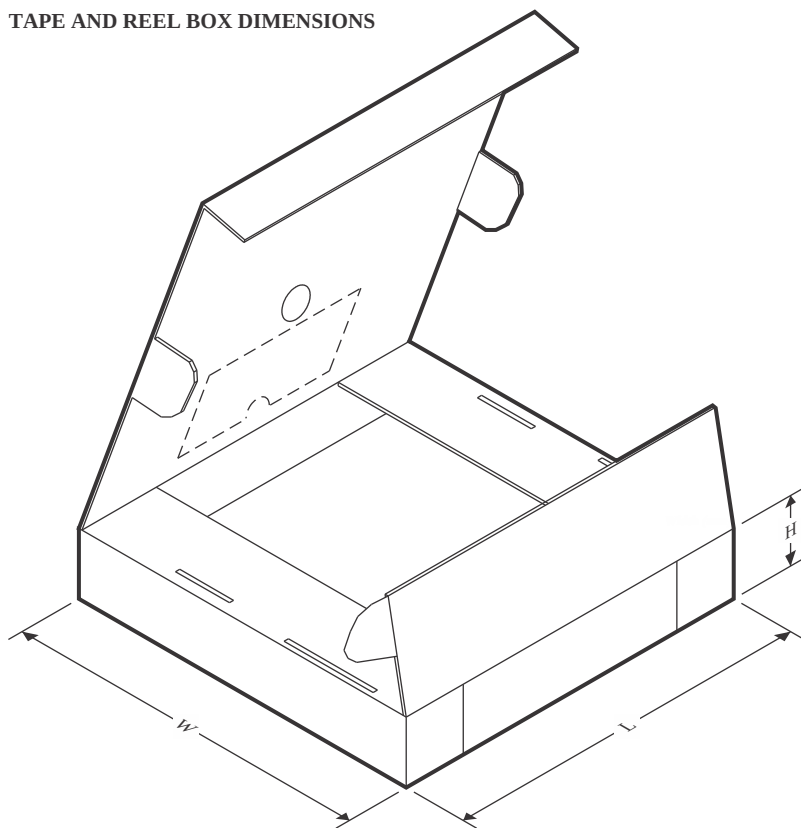


\*All dimensions are nominal

| Device     | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| TL061ACDR  | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL061CDR   | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL061CPSR  | SO           | PS              | 8    | 2000 | 330.0              | 16.4               | 8.35    | 6.6     | 2.4     | 12.0    | 16.0   | Q1            |
| TL061IDR   | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL062ACDR  | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL062ACPSR | SO           | PS              | 8    | 2000 | 330.0              | 16.4               | 8.35    | 6.6     | 2.4     | 12.0    | 16.0   | Q1            |
| TL062BCDR  | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL062CDR   | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL062CPSR  | SO           | PS              | 8    | 2000 | 330.0              | 16.4               | 8.35    | 6.6     | 2.4     | 12.0    | 16.0   | Q1            |
| TL062CPWR  | TSSOP        | PW              | 8    | 2000 | 330.0              | 12.4               | 7.0     | 3.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL062IDR   | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL062IPWR  | TSSOP        | PW              | 8    | 2000 | 330.0              | 12.4               | 7.0     | 3.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL064ACDR  | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL064BCDR  | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL064CDR   | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL064CDR   | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |

| Device      | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|-------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| TL064CDR    | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL064CNSR   | SOP          | NS              | 14   | 2000 | 330.0              | 16.4               | 8.1     | 10.4    | 2.5     | 12.0    | 16.0   | Q1            |
| TL064CPWR   | TSSOP        | PW              | 14   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL064CPWRG4 | TSSOP        | PW              | 14   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL064CPWRG4 | TSSOP        | PW              | 14   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL064IDR    | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL064INSR   | SOP          | NS              | 14   | 2000 | 330.0              | 16.4               | 8.1     | 10.4    | 2.5     | 12.0    | 16.0   | Q1            |
| TL064IPWR   | TSSOP        | PW              | 14   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL064IPWR   | TSSOP        | PW              | 14   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |

## TAPE AND REEL BOX DIMENSIONS

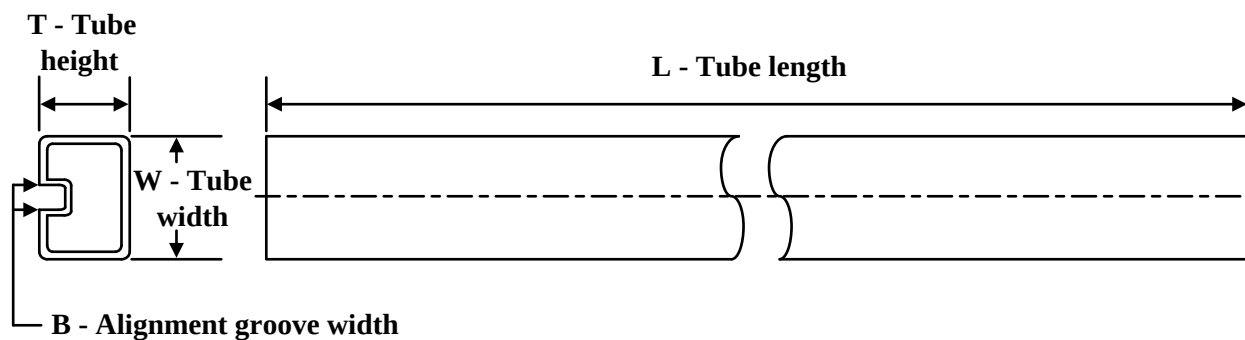


\*All dimensions are nominal

| Device     | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|------------|--------------|-----------------|------|------|-------------|------------|-------------|
| TL061ACDR  | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL061CDR   | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL061CPSR  | SO           | PS              | 8    | 2000 | 353.0       | 353.0      | 32.0        |
| TL061IDR   | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL062ACDR  | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL062ACPSR | SO           | PS              | 8    | 2000 | 353.0       | 353.0      | 32.0        |
| TL062BCDR  | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL062CDR   | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL062CPSR  | SO           | PS              | 8    | 2000 | 353.0       | 353.0      | 32.0        |
| TL062CPWR  | TSSOP        | PW              | 8    | 2000 | 353.0       | 353.0      | 32.0        |
| TL062IDR   | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL062IPWR  | TSSOP        | PW              | 8    | 2000 | 353.0       | 353.0      | 32.0        |
| TL064ACDR  | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL064BCDR  | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL064CDR   | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL064CDR   | SOIC         | D               | 14   | 2500 | 333.2       | 345.9      | 28.6        |
| TL064CDR   | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL064CNSR  | SOP          | NS              | 14   | 2000 | 353.0       | 353.0      | 32.0        |

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| Device      | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|-------------|--------------|-----------------|------|------|-------------|------------|-------------|
| TL064CPWR   | TSSOP        | PW              | 14   | 2000 | 356.0       | 356.0      | 35.0        |
| TL064CPWRG4 | TSSOP        | PW              | 14   | 2000 | 356.0       | 356.0      | 35.0        |
| TL064CPWRG4 | TSSOP        | PW              | 14   | 2000 | 356.0       | 356.0      | 35.0        |
| TL064IDR    | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL064INSR   | SOP          | NS              | 14   | 2000 | 353.0       | 353.0      | 32.0        |
| TL064IPWR   | TSSOP        | PW              | 14   | 2000 | 353.0       | 353.0      | 32.0        |
| TL064IPWR   | TSSOP        | PW              | 14   | 2000 | 353.0       | 353.0      | 32.0        |

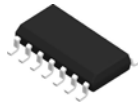
**TUBE**


\*All dimensions are nominal

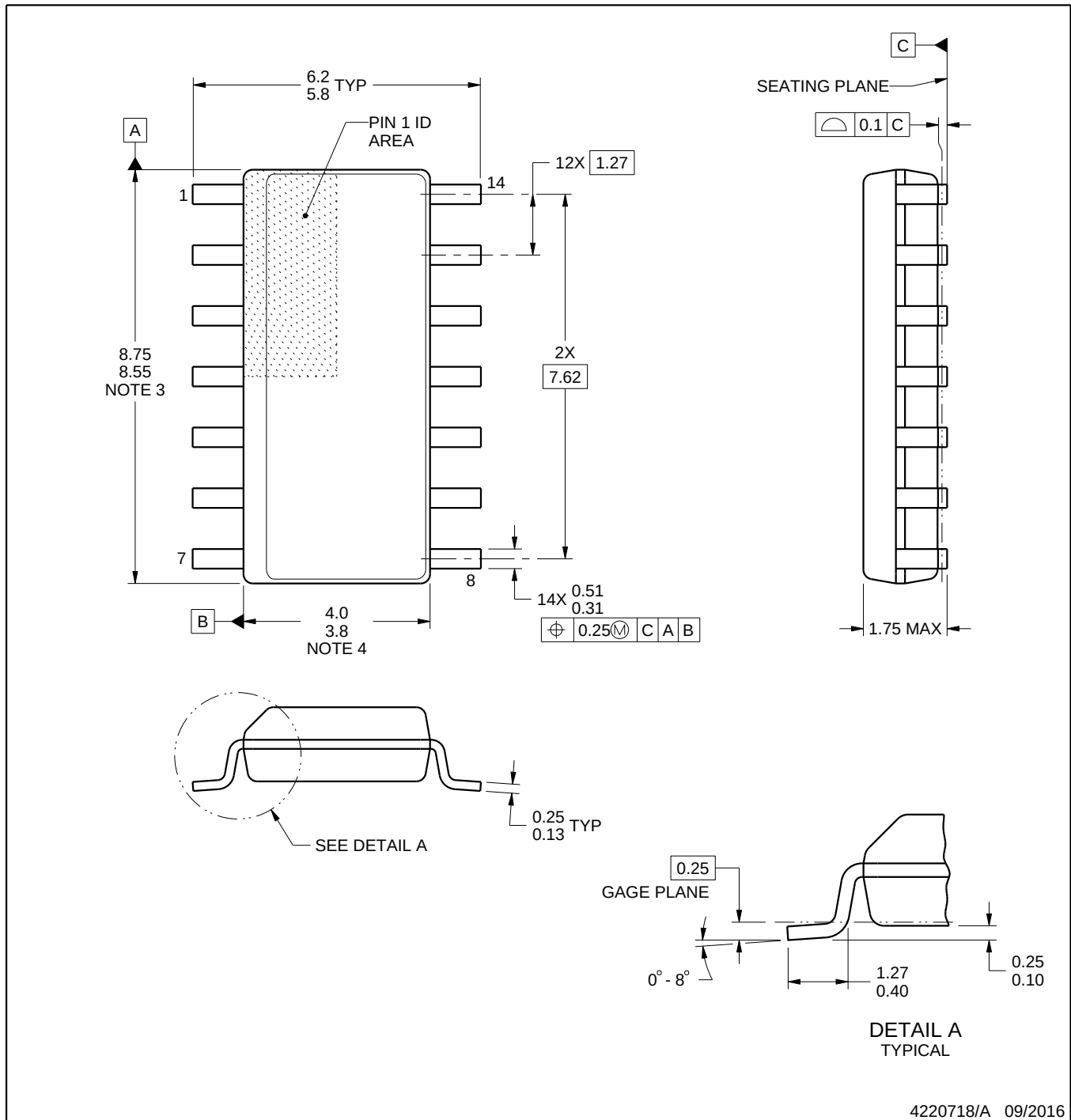
| Device      | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (μm) | B (mm) |
|-------------|--------------|--------------|------|-----|--------|--------|--------|--------|
| 81023022A   | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| 81023032A   | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| 8102303DA   | W            | CFP          | 14   | 25  | 506.98 | 26.16  | 6220   | NA     |
| TL061ACP    | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL061ACP.A  | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL061BCP    | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL061BCP.A  | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL061CP     | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL061CP.A   | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL061IP     | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL061IP.A   | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL062ACP    | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL062ACP.A  | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL062BCP    | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL062BCP.A  | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL062CP     | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL062CP.A   | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL062CPS    | PS           | SOP          | 8    | 80  | 530    | 10.5   | 4000   | 4.1    |
| TL062CPS.A  | PS           | SOP          | 8    | 80  | 530    | 10.5   | 4000   | 4.1    |
| TL062IP     | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL062IP.A   | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL062MFKB   | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| TL062MFKB.A | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| TL064ACN    | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL064ACN.A  | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL064BCN    | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL064BCN.A  | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL064CN     | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL064CN.A   | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |



| Device      | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (μm) | B (mm) |
|-------------|--------------|--------------|------|-----|--------|--------|--------|--------|
| TL064IN     | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL064IN.A   | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL064INS    | NS           | SOP          | 14   | 50  | 530    | 10.5   | 4000   | 4.1    |
| TL064INS.A  | NS           | SOP          | 14   | 50  | 530    | 10.5   | 4000   | 4.1    |
| TL064MFKB   | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| TL064MFKB.A | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| TL064MWB    | W            | CFP          | 14   | 25  | 506.98 | 26.16  | 6220   | NA     |
| TL064MWB.A  | W            | CFP          | 14   | 25  | 506.98 | 26.16  | 6220   | NA     |

**D0014A****PACKAGE OUTLINE****SOIC - 1.75 mm max height**

SMALL OUTLINE INTEGRATED CIRCUIT

**NOTES:**

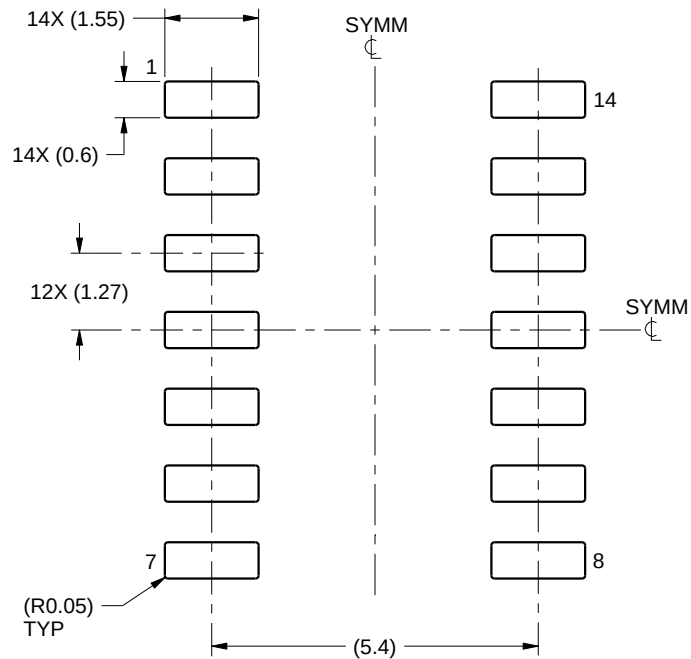
1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm, per side.
5. Reference JEDEC registration MS-012, variation AB.

# EXAMPLE BOARD LAYOUT

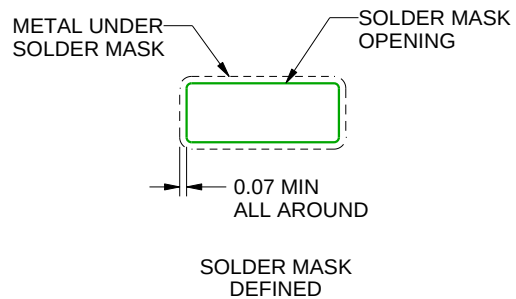
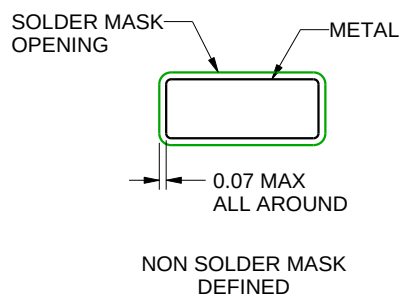
D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



LAND PATTERN EXAMPLE  
SCALE:8X



SOLDER MASK DETAILS

4220718/A 09/2016

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

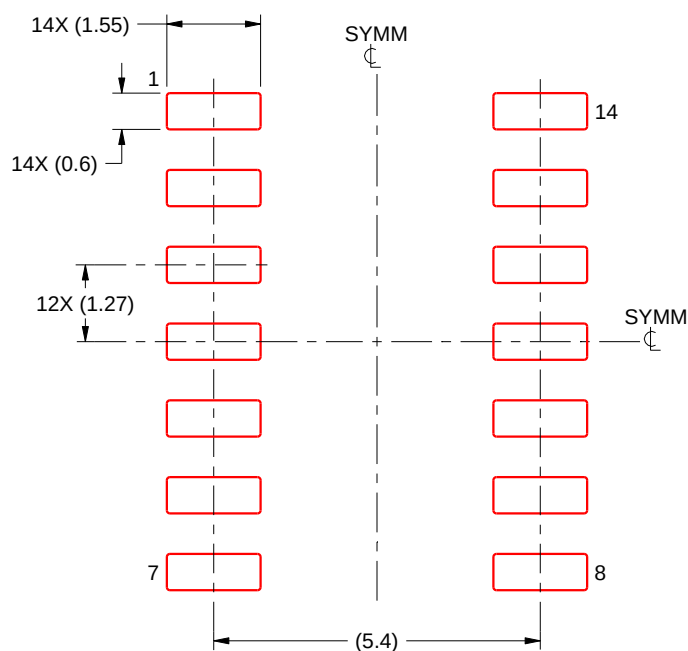
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE:8X

4220718/A 09/2016

NOTES: (continued)

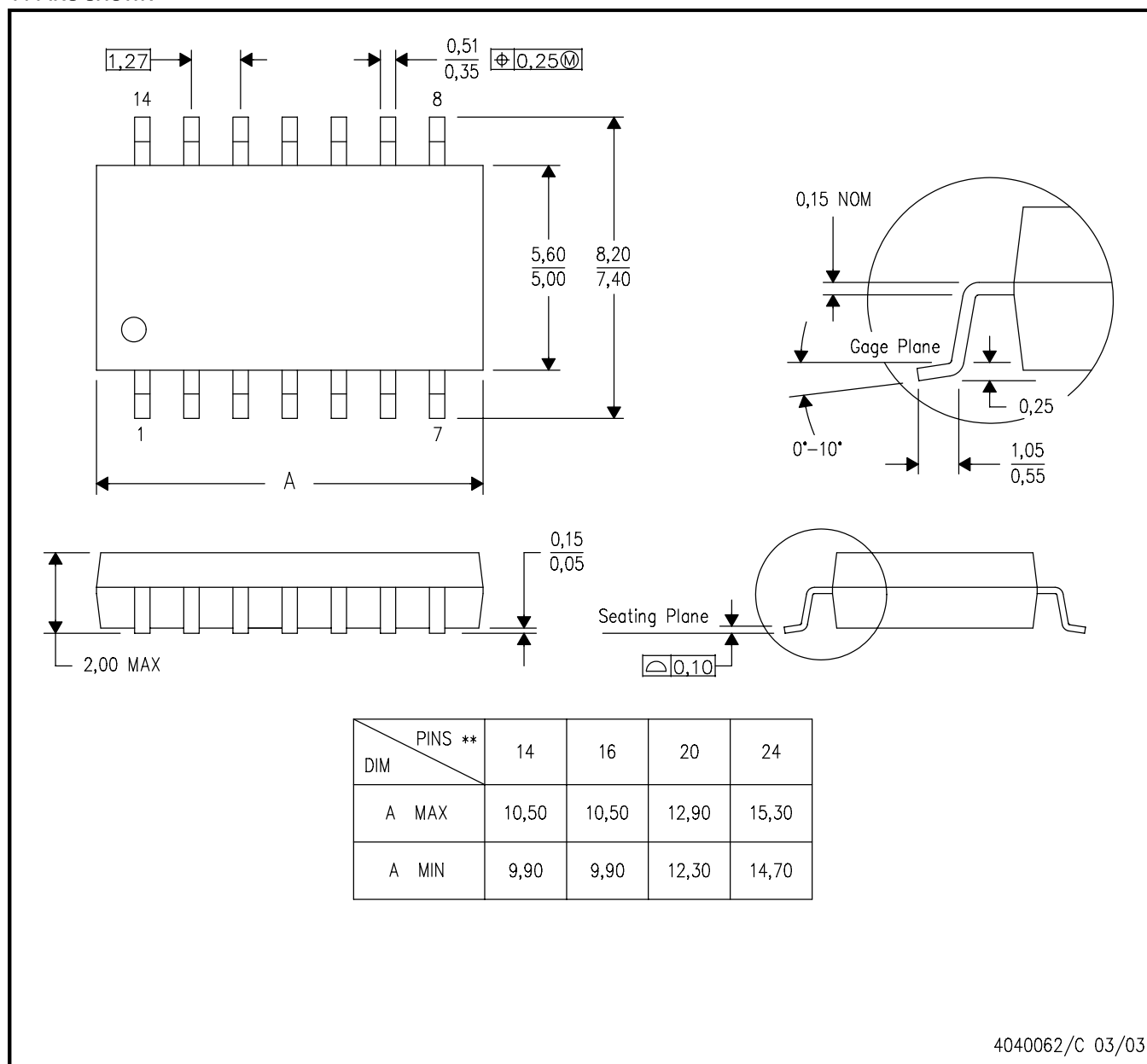
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

# MECHANICAL DATA

NS (R-PDSO-G\*\*)

PLASTIC SMALL-OUTLINE PACKAGE

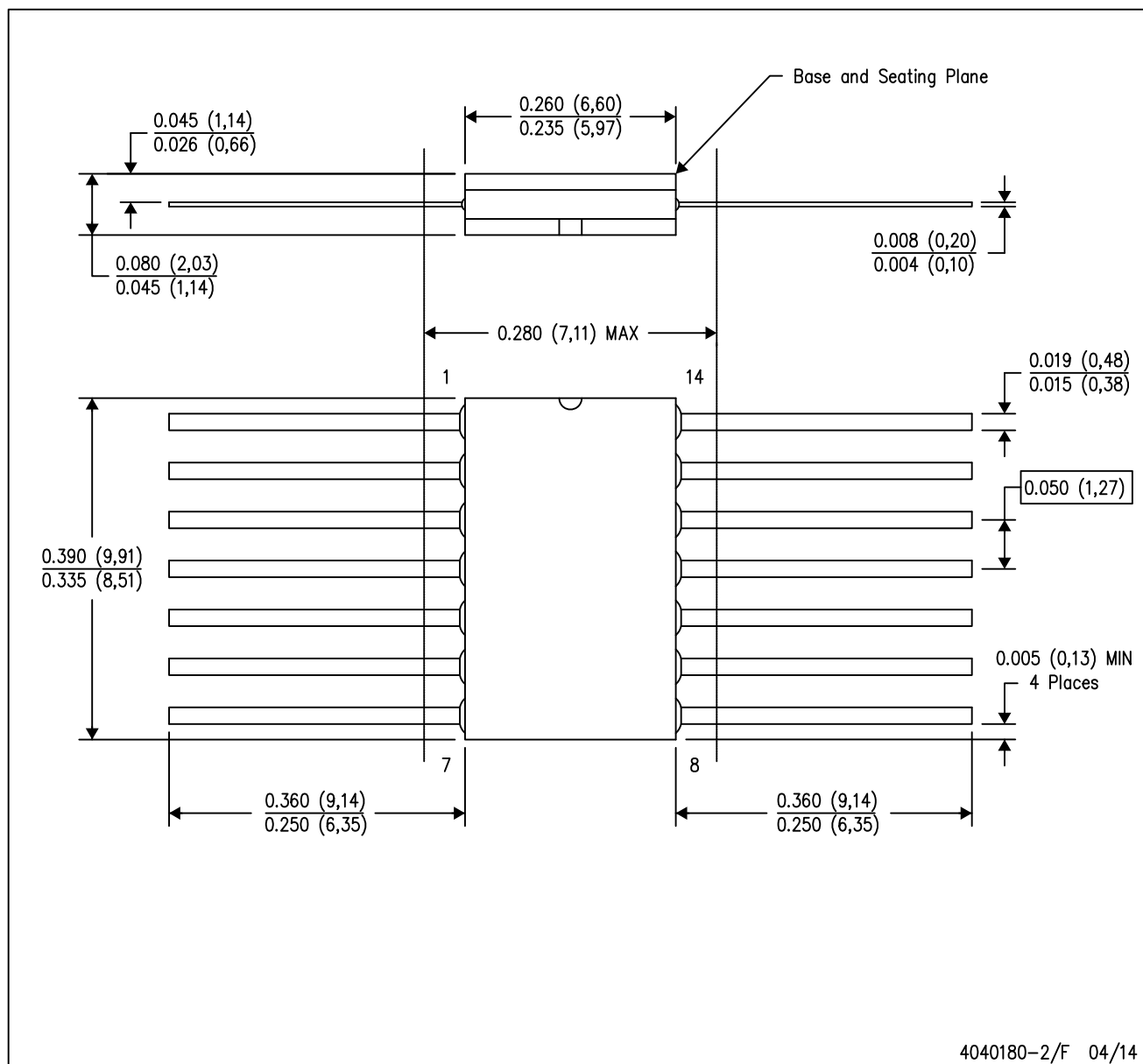
14-PINS SHOWN



- NOTES:
- A. All linear dimensions are in millimeters.
  - B. This drawing is subject to change without notice.
  - C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

W (R-GDFP-F14)

CERAMIC DUAL FLATPACK



## GENERIC PACKAGE VIEW

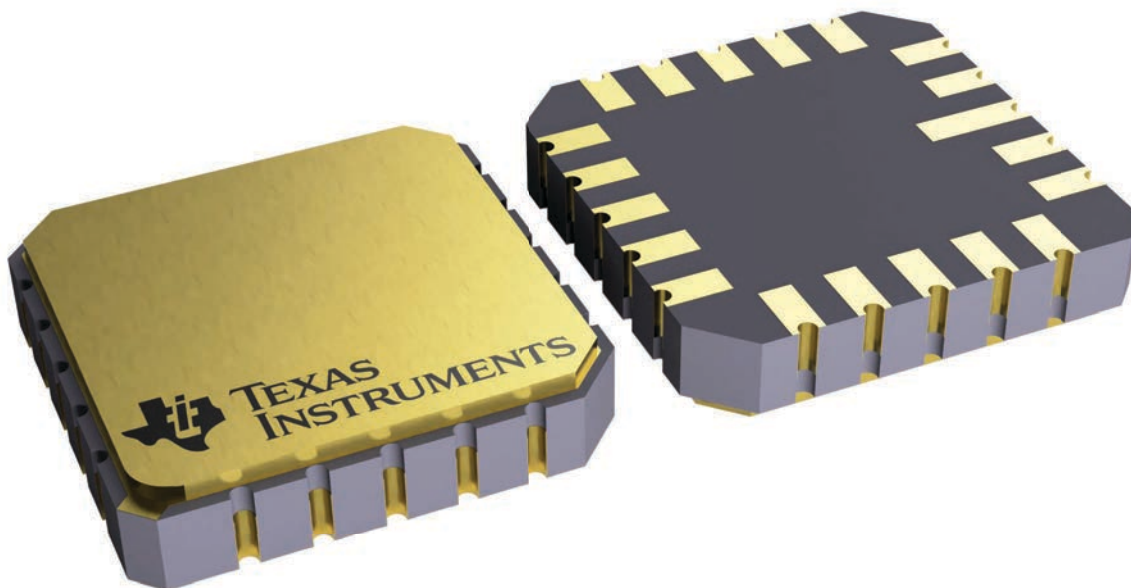
**FK 20**

**LCCC - 2.03 mm max height**

8.89 x 8.89, 1.27 mm pitch

LEADLESS CERAMIC CHIP CARRIER

This image is a representation of the package family, actual package may vary.  
Refer to the product data sheet for package details.



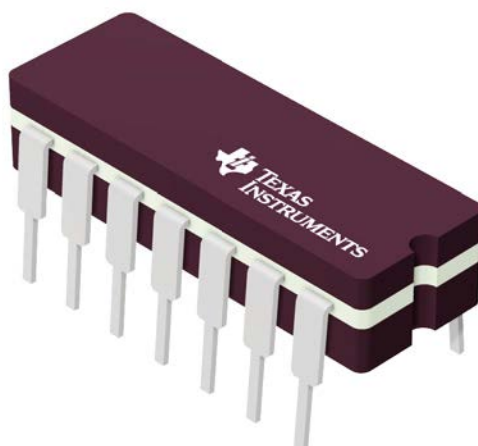
4229370VA\

**J 14**

## GENERIC PACKAGE VIEW

**CDIP - 5.08 mm max height**

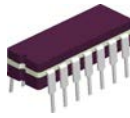
CERAMIC DUAL IN LINE PACKAGE



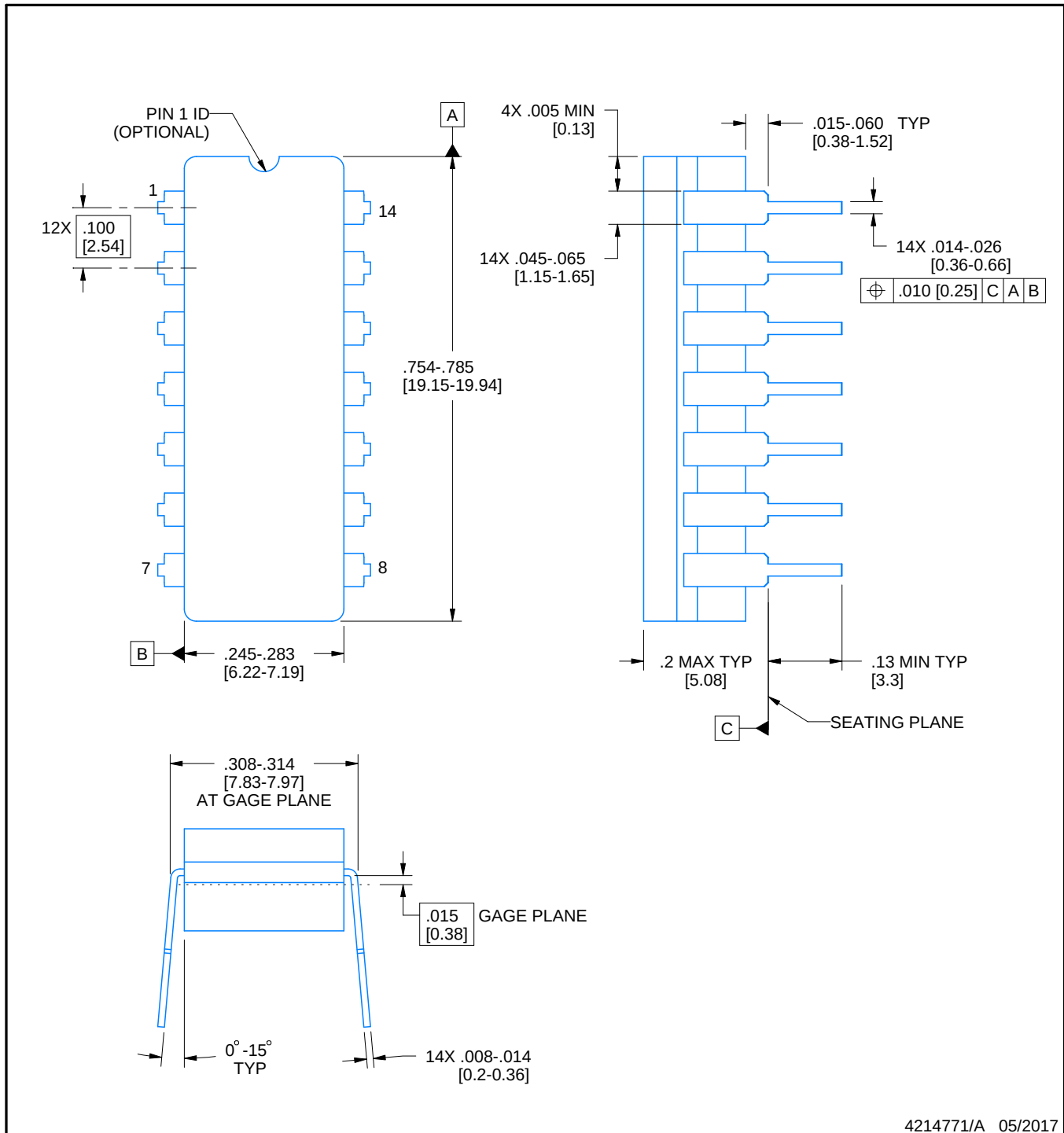
Images above are just a representation of the package family, actual package may vary.  
Refer to the product data sheet for package details.

4040083-5/G



**J0014A****PACKAGE OUTLINE****CDIP - 5.08 mm max height**

CERAMIC DUAL IN LINE PACKAGE



4214771/A 05/2017

**NOTES:**

1. All controlling linear dimensions are in inches. Dimensions in brackets are in millimeters. Any dimension in brackets or parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This package is hermetically sealed with a ceramic lid using glass frit.
4. Index point is provided on cap for terminal identification only and on press ceramic glass frit seal only.
5. Falls within MIL-STD-1835 and GDIP1-T14.



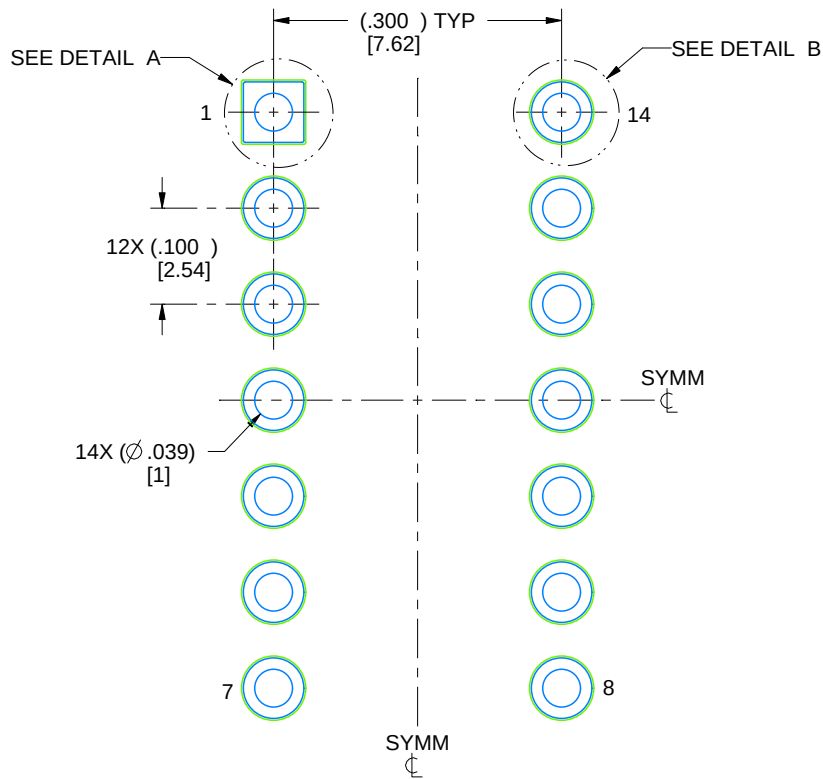
**TEXAS  
INSTRUMENTS**  
www.ti.com

# EXAMPLE BOARD LAYOUT

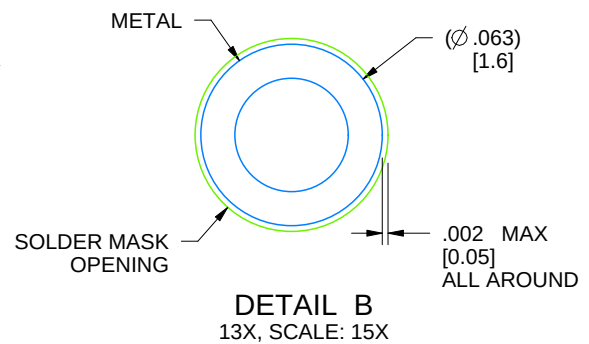
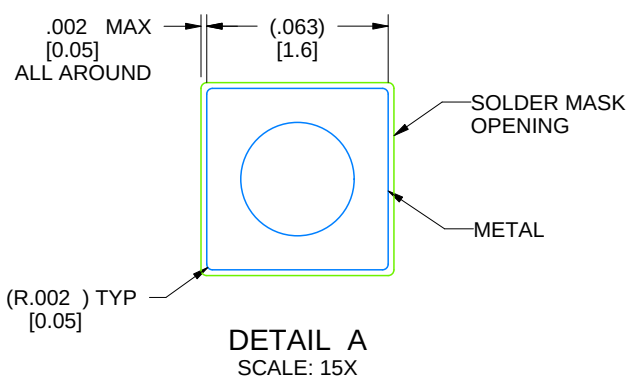
J0014A

CDIP - 5.08 mm max height

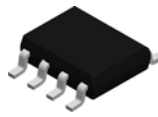
CERAMIC DUAL IN LINE PACKAGE



LAND PATTERN EXAMPLE  
NON-SOLDER MASK DEFINED  
SCALE: 5X



4214771/A 05/2017

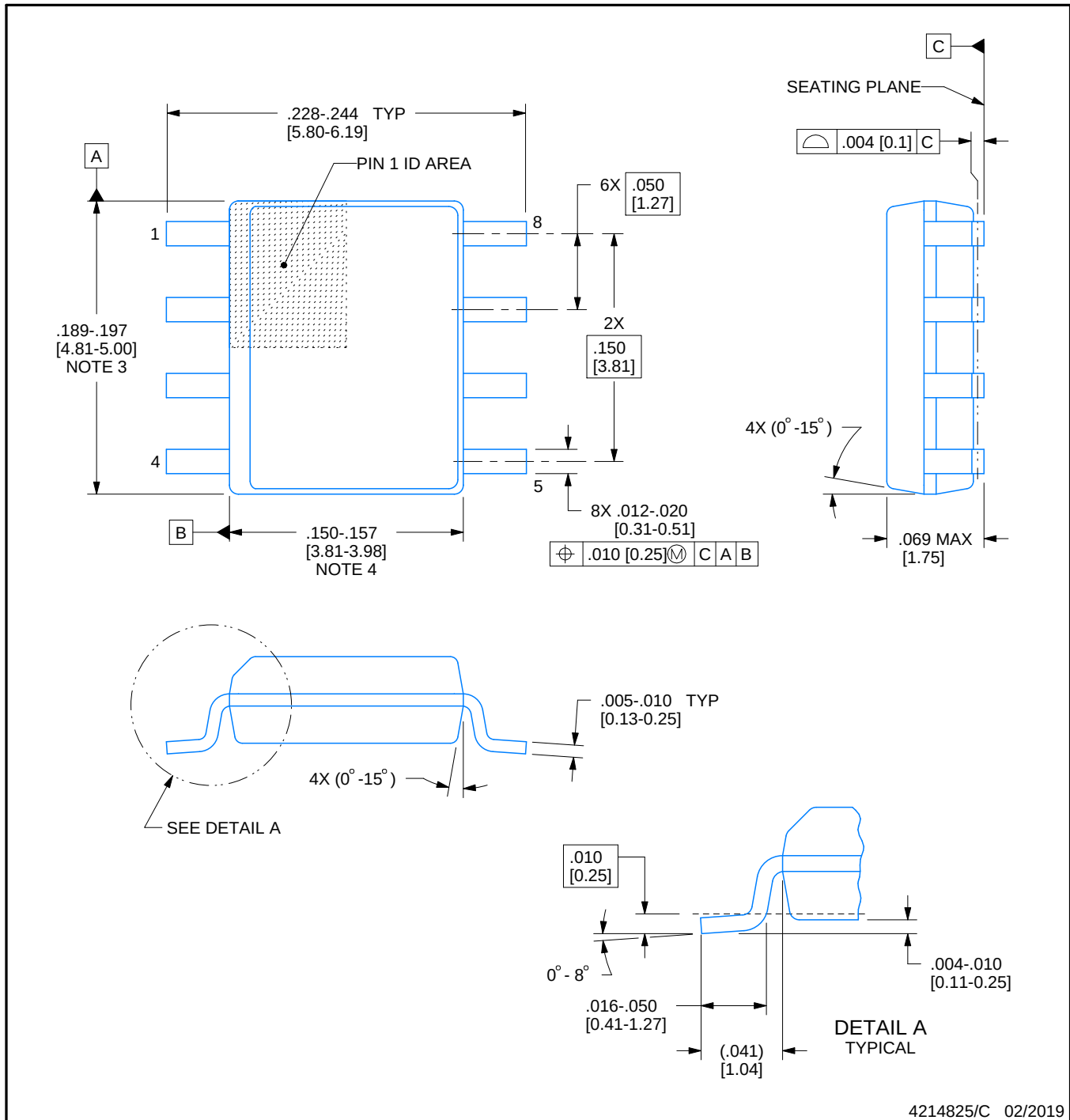


**D0008A**

# PACKAGE OUTLINE

**SOIC - 1.75 mm max height**

SMALL OUTLINE INTEGRATED CIRCUIT



4214825/C 02/2019

## NOTES:

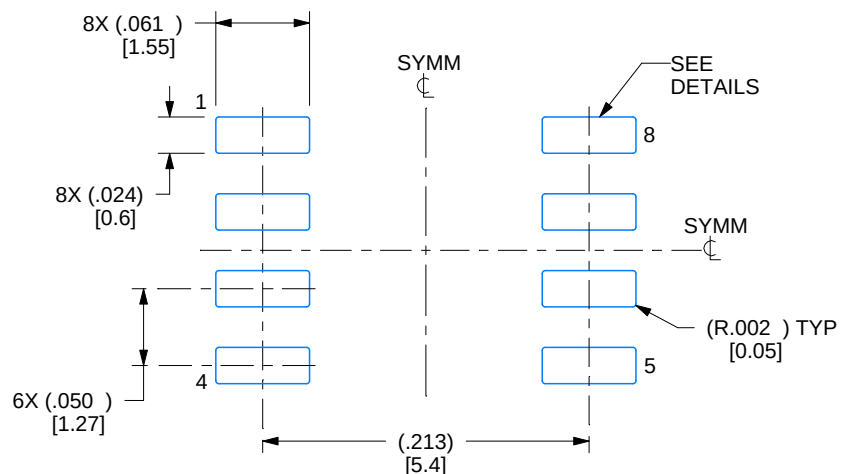
1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.

# EXAMPLE BOARD LAYOUT

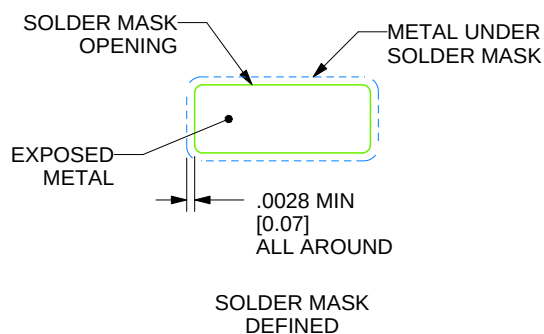
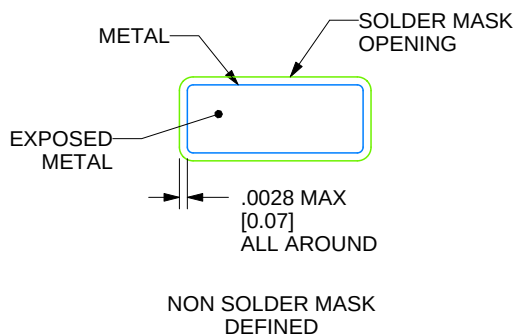
D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE:8X



SOLDER MASK DETAILS

4214825/C 02/2019

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

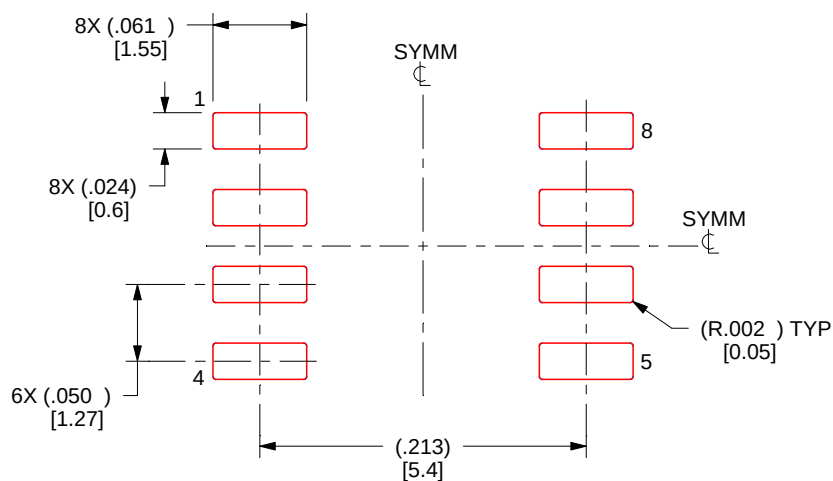
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE  
BASED ON .005 INCH [0.125 MM] THICK STENCIL  
SCALE:8X

4214825/C 02/2019

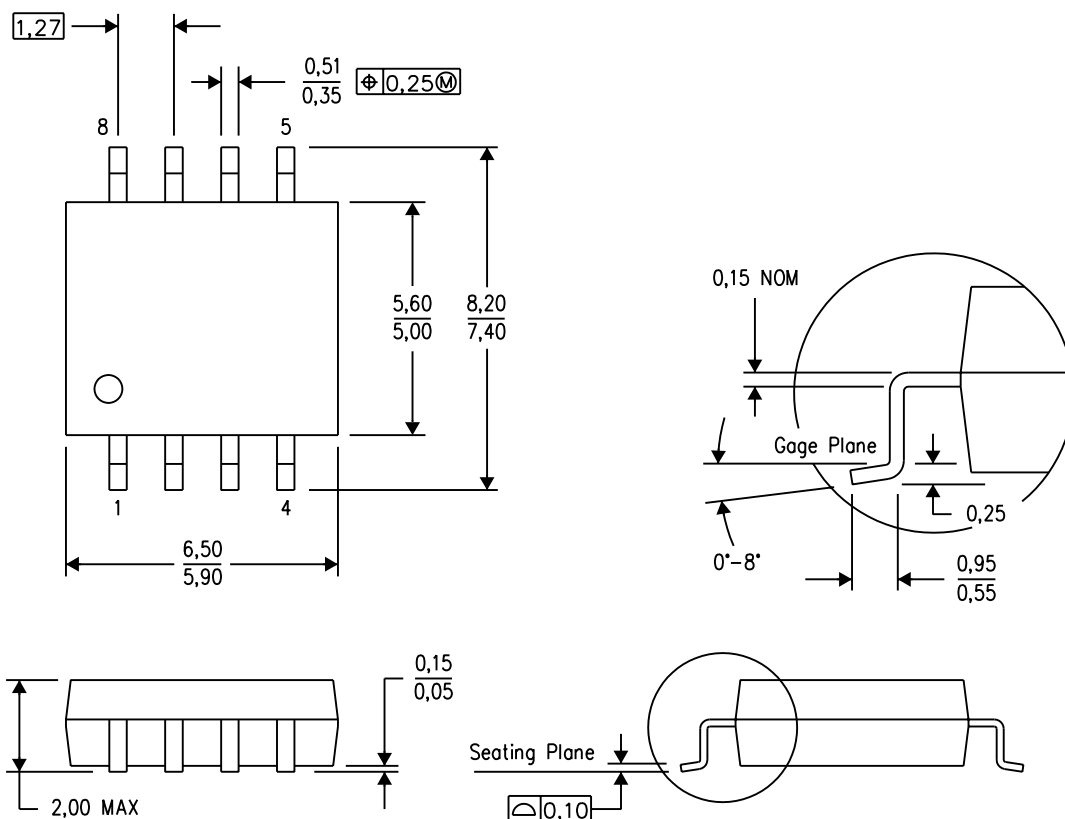
NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

## MECHANICAL DATA

PS (R-PDSO-G8)

PLASTIC SMALL-OUTLINE PACKAGE

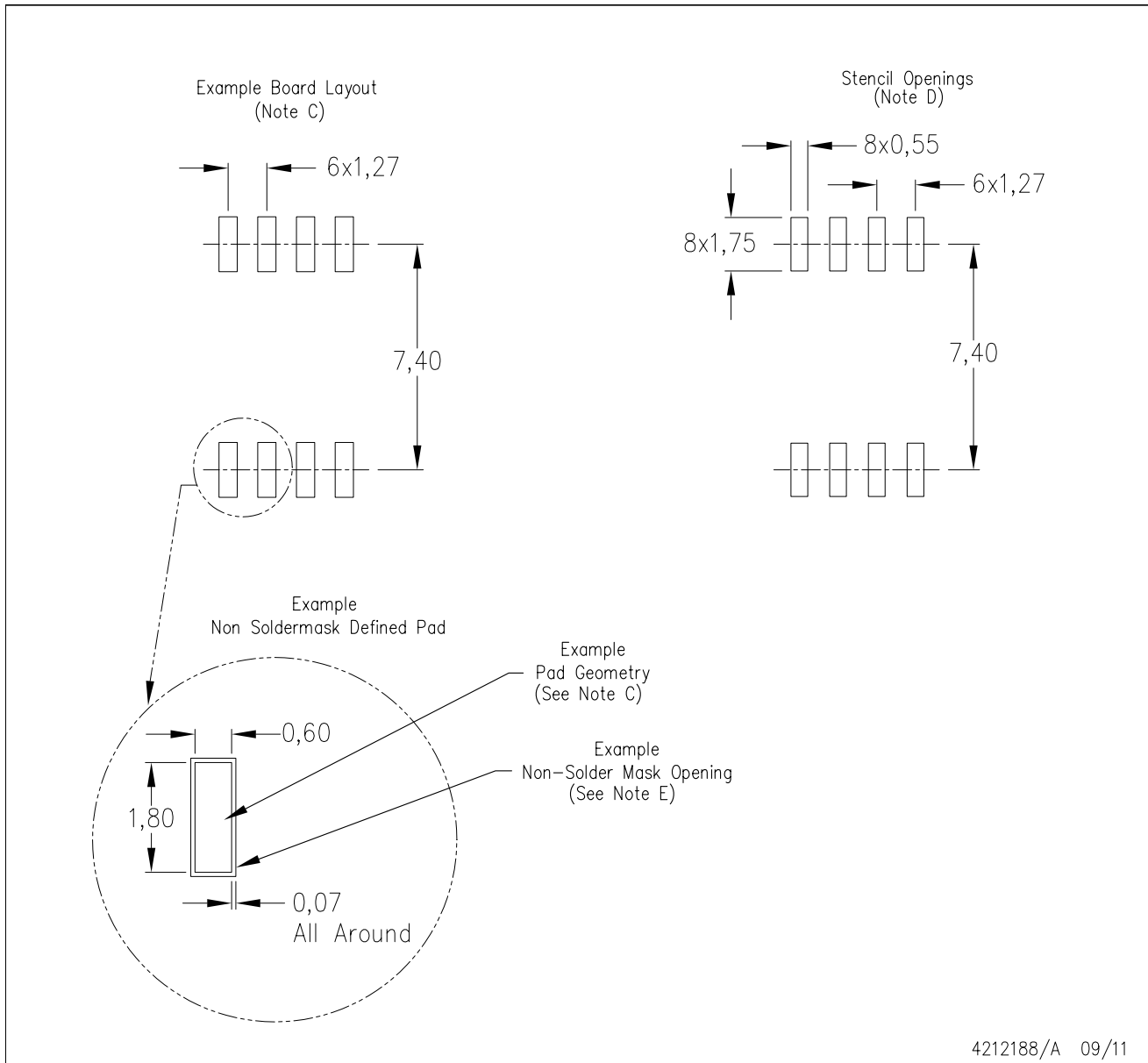


4040063/C 03/03

- NOTES:
- A. All linear dimensions are in millimeters.
  - B. This drawing is subject to change without notice.
  - C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

## PS (R-PDSO-G8)

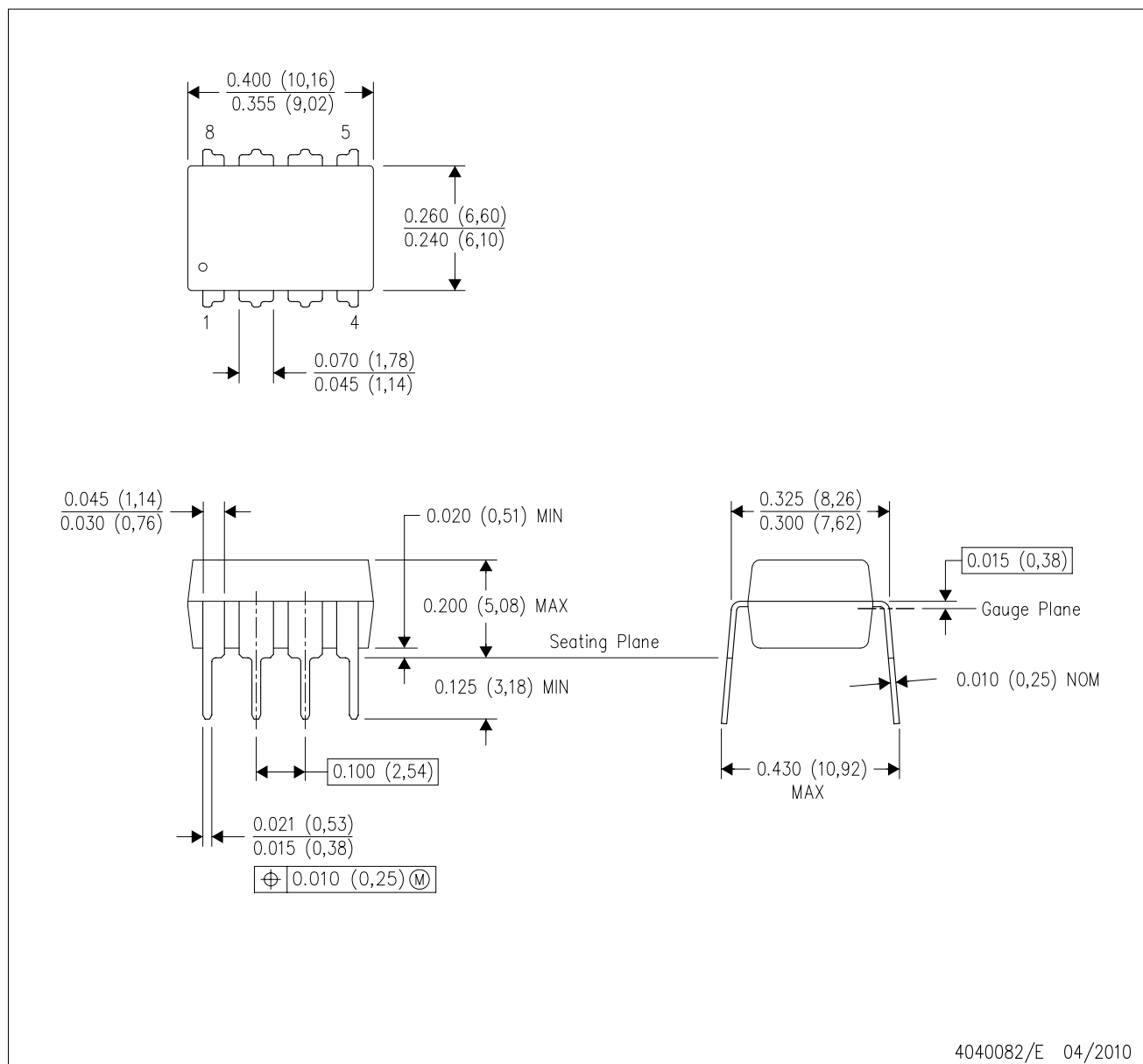
## PLASTIC SMALL OUTLINE



- NOTES:
- All linear dimensions are in millimeters.
  - This drawing is subject to change without notice.
  - Publication IPC-7351 is recommended for alternate designs.
  - Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC-7525 for other stencil recommendations.
  - Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

P (R-PDIP-T8)

PLASTIC DUAL-IN-LINE PACKAGE



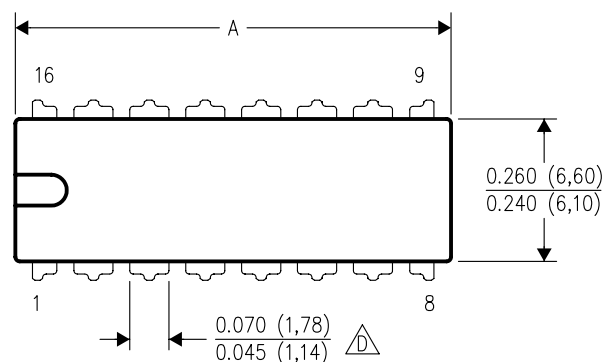
- NOTES:
- A. All linear dimensions are in inches (millimeters).
  - B. This drawing is subject to change without notice.
  - C. Falls within JEDEC MS-001 variation BA.



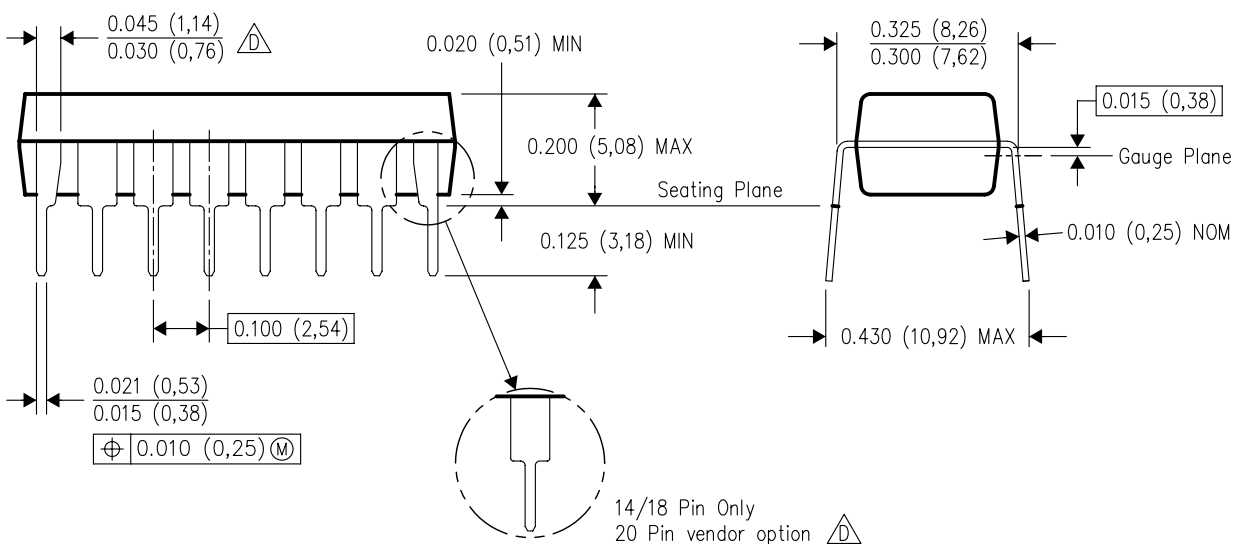
N (R-PDIP-T\*\*)

16 PINS SHOWN

## PLASTIC DUAL-IN-LINE PACKAGE

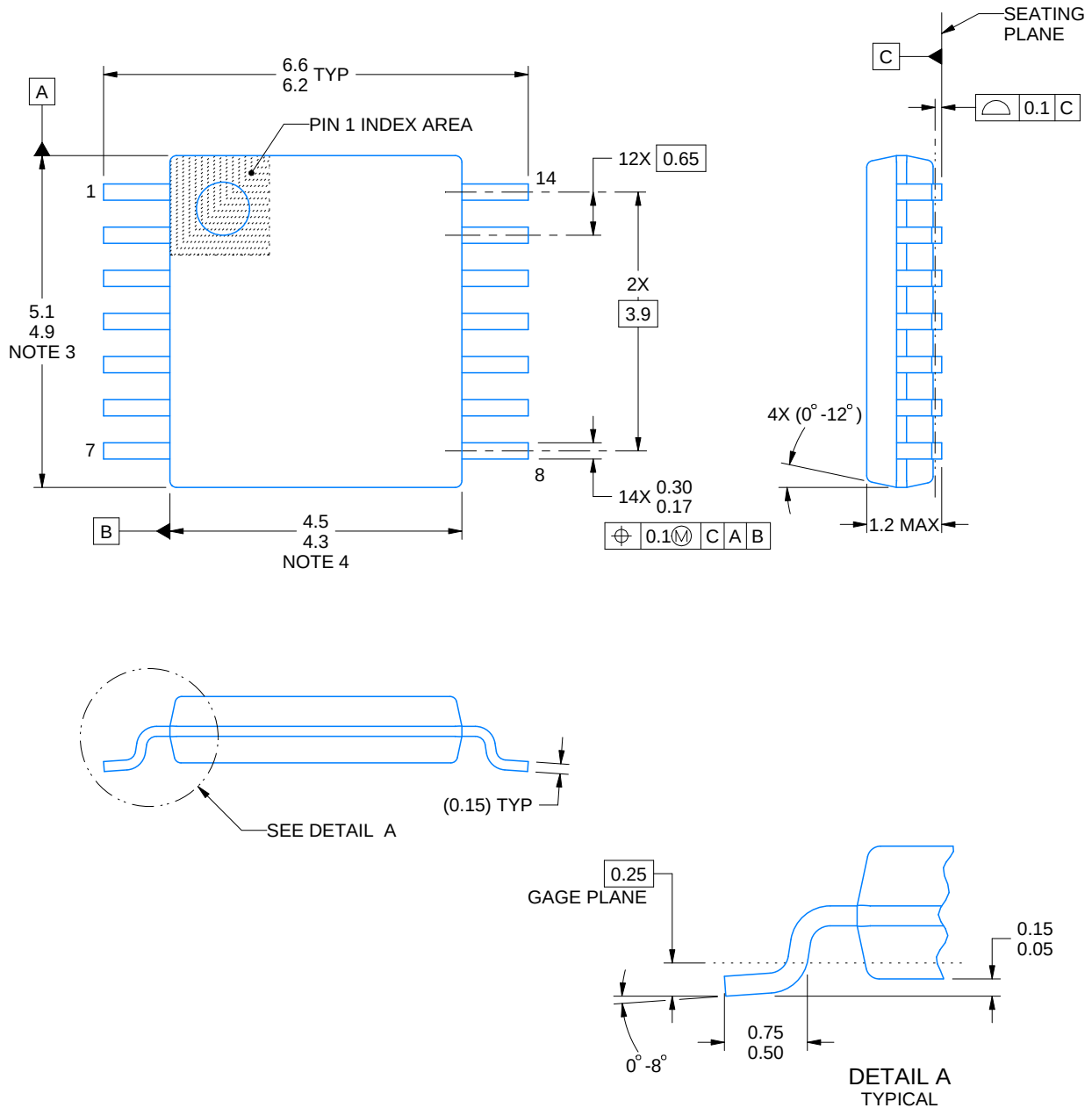
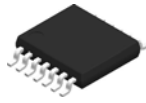


| PINS **<br>DIM      | 14               | 16               | 18               | 20               |
|---------------------|------------------|------------------|------------------|------------------|
| A MAX               | 0.775<br>(19,69) | 0.775<br>(19,69) | 0.920<br>(23,37) | 1.060<br>(26,92) |
| A MIN               | 0.745<br>(18,92) | 0.745<br>(18,92) | 0.850<br>(21,59) | 0.940<br>(23,88) |
| MS-001<br>VARIATION | AA               | BB               | AC               | AD               |



4040049/E 12/2002

- NOTES:
- A. All linear dimensions are in inches (millimeters).
  - B. This drawing is subject to change without notice.
  -  Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
  -  The 20 pin end lead shoulder width is a vendor option, either half or full width.



4220202/B 12/2023

## NOTES:

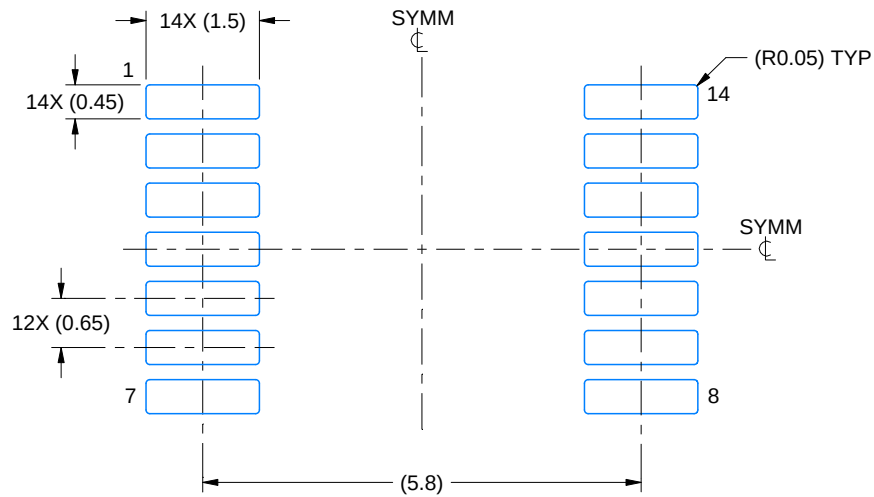
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.

# EXAMPLE BOARD LAYOUT

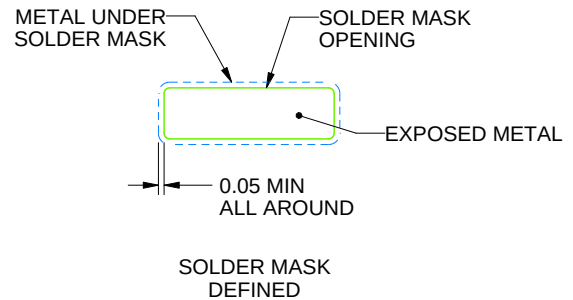
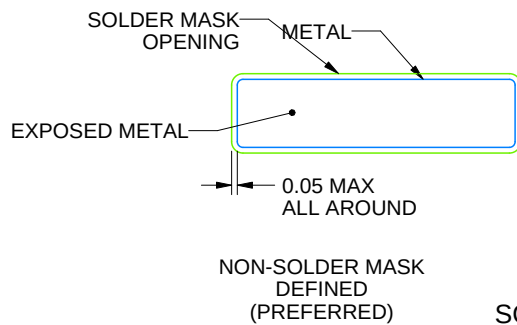
PW0014A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE: 10X



SOLDER MASK DETAILS

4220202/B 12/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

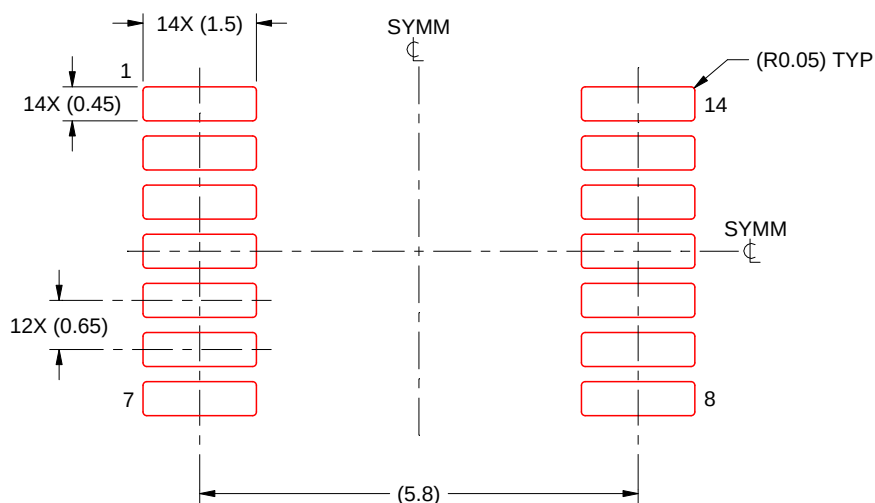
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

# EXAMPLE STENCIL DESIGN

PW0014A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE: 10X

4220202/B 12/2023

NOTES: (continued)

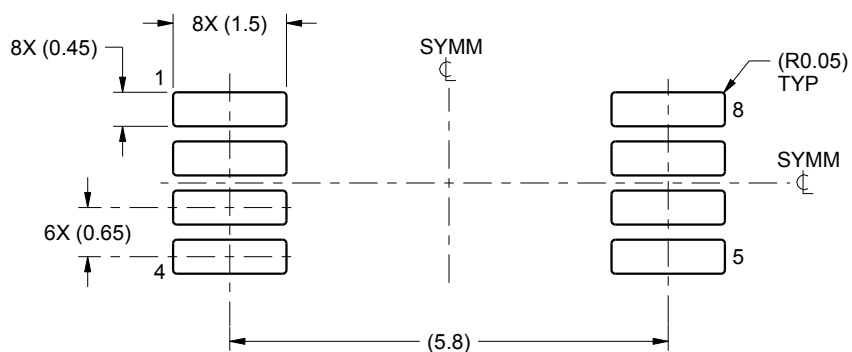
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.



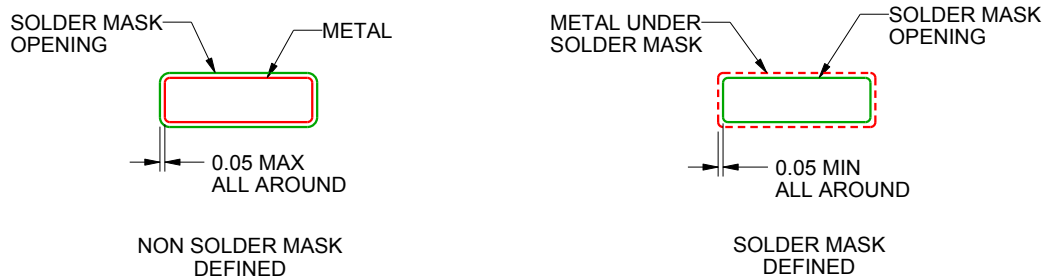
**PW0008A**

## TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
SCALE:10X



SOLDER MASK DETAILS  
NOT TO SCALE

4221848/A 02/2015

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

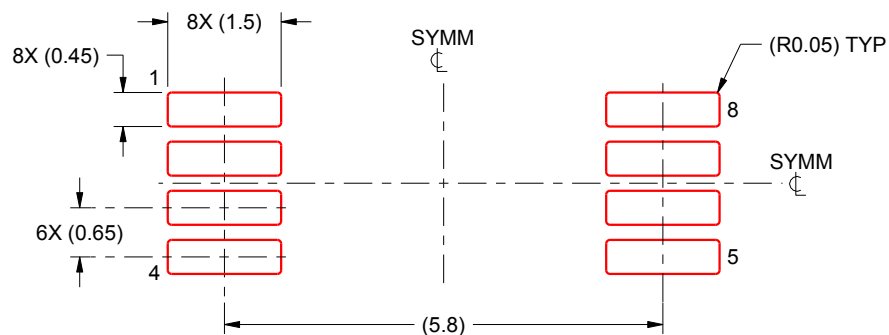
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

PW0008A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE:10X

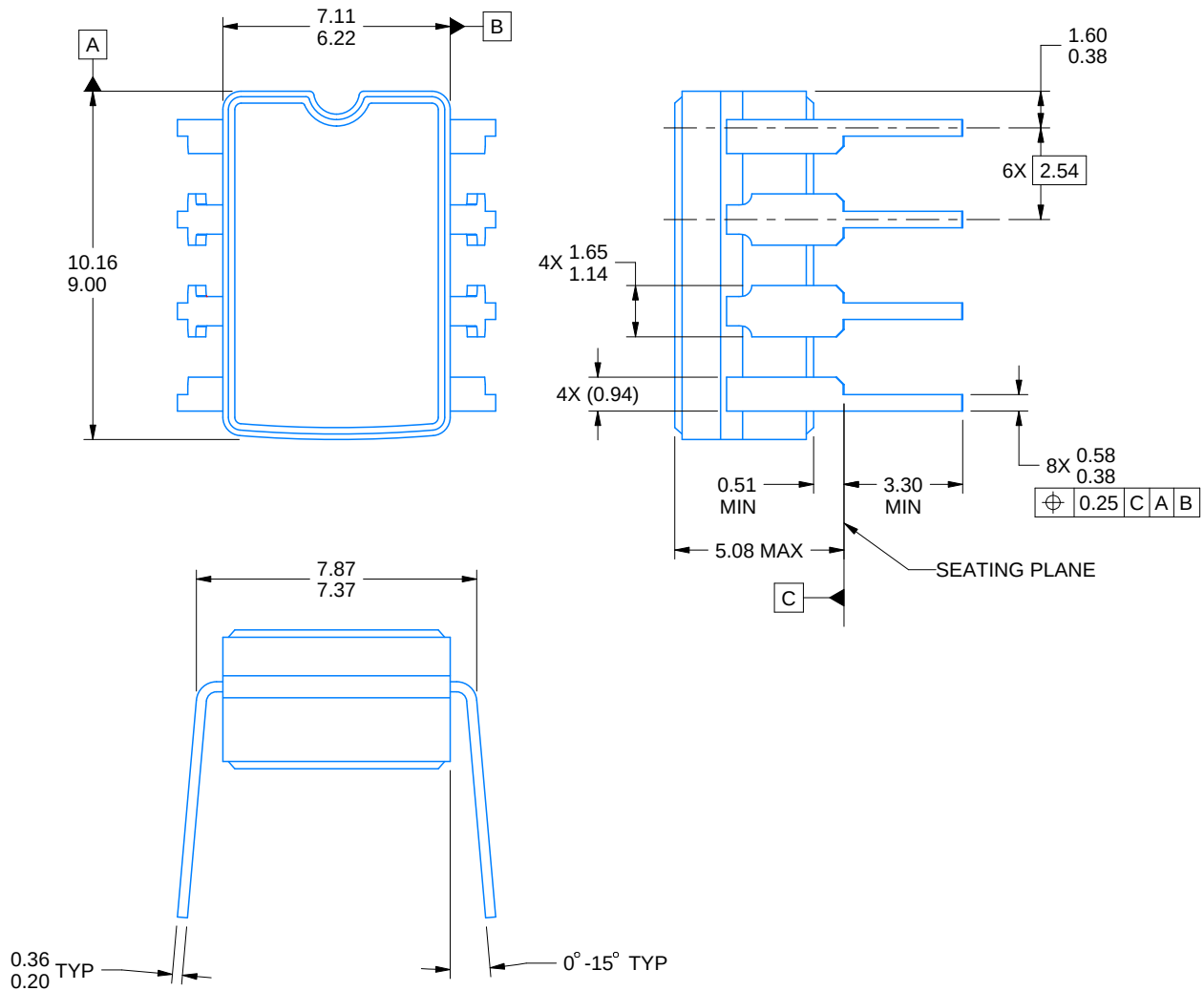
4221848/A 02/2015

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

**JG0008A****PACKAGE OUTLINE****CDIP - 5.08 mm max height**

CERAMIC DUAL IN-LINE PACKAGE



4230036/A 09/2023

**NOTES:**

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This package can be hermetically sealed with a ceramic lid using glass frit.
4. Index point is provided on cap for terminal identification.
5. Falls within MIL STD 1835 GDIP1-T8

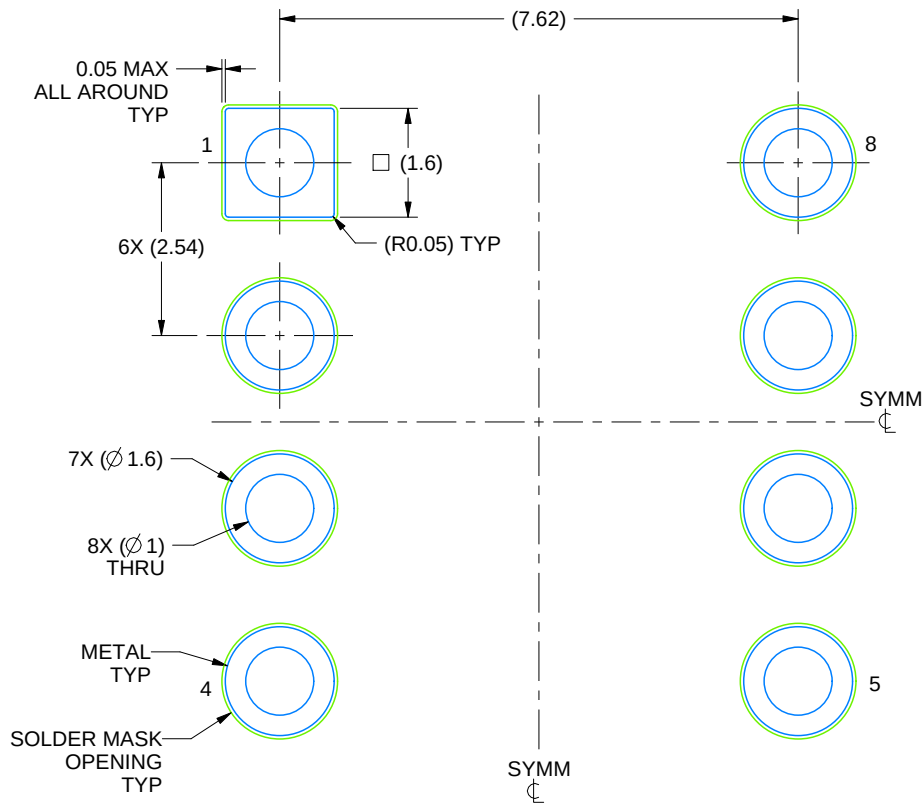


# EXAMPLE BOARD LAYOUT

JG0008A

CDIP - 5.08 mm max height

CERAMIC DUAL IN-LINE PACKAGE



LAND PATTERN EXAMPLE  
NON SOLDER MASK DEFINED  
SCALE: 9X

4230036/A 09/2023

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